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Optimal Prediction of Bitcoin High and Low Prices: An Exploratory Analysis

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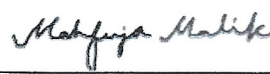
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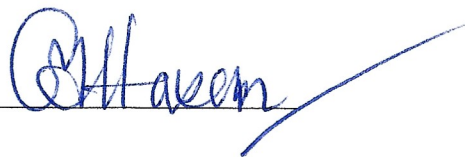
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By Tatiana Rice

Abstract

The purpose of this paper is to provide traders with a trader friendly model that would enable them to accurately predict Bitcoin's high price and low price so that they are able to make more informed decisions for improved risk management when trading the highly volatile asset – Bitcoin. To achieve this purpose, this paper poses the following research question: Which statistical model-frequency combination best predicts – in terms of Mean Absolute Percent Error (MAPE), Akaike Information Criterion (AIC), and Schwartz Information Criterion (SIC) – Bitcoin's high price and low price? This paper also poses the objective of ensuring that the research question is both answered and that the statistical model-frequency combination selected is trader friendly or in other words, both easy to implement and interpret. To answer the research question and to meet the objectives of this paper, this paper develops two types of statistical models – Ordinary Least Squares (OLS) and Autoregressive Distributed Lags (ARDL). These models are developed across multiple frequencies – daily, weekly, and monthly. These statistical models and frequencies are chosen due to their trader friendliness, which aligns with the objective of this paper. Results for each of these models training sets show that the ARDL Bull Weekly model best predicts Bitcoin's high price while the ARDL Bull Monthly model best predicts Bitcoin's low price. To test the robustness of these two models, each model is tested on 7 robustness tests that incorporate out of sample data that covers various market conditions from 2011 to 2024. Results for the robustness tests show that both the ARDL Bull Weekly model and ARDL Bull Monthly model are highly robust for the out of sample data and market conditions they are tested on. This is evident in both models' remarkable average mean absolute percentage error (MAPE) across out of sample robustness tests of 1.3% (ARDL Bull Weekly) and 3.28% (ARDL Bull Monthly). This indicates that these models could be used by traders to accurately predict Bitcoin's high price and low price so that they are able to make more informed decisions for improved risk management.

JEL Classification: C01, C02, C52, C53, C58

Keywords: Bitcoin Price Forecasting, Bitcoin Trading, Statistical Modeling, Ordinary Least Squares, Autoregressive Distributed Lags, Regime Dependent Analysis, Robustness Testing, Risk Management

Introduction

The primary question addressed in this paper is: Which statistical model-frequency combination best predicts – in terms of Mean Absolute Percent Error (MAPE), Akaike Information Criterion (AIC), and Schwartz Information Criterion (SIC) – Bitcoin’s high and low prices? The main objective is to answer the above primary question, with a secondary goal of ensuring that the selected combination is trader-friendly—easy to implement and interpret. These objectives are crucial for two reasons. First, Bitcoin’s high volatility necessitates accurate forecasting tools to help traders make informed decisions for improved risk management. Second, the majority of cryptocurrency traders are retail traders with basic to intermediate knowledge of forecasting. A trader-friendly model reduces the likelihood of errors in implementation and interpretation, which could negatively impact risk management.

This paper selects Bitcoin’s high and low prices as the dependent variables for one major reason – optimizing forecasts for Bitcoin’s high and low prices, which are underexplored within the literature, rather than other more commonly used variables like Bitcoin’s spot price, enhances risk management by allowing traders to set precise entry and exit points.

This paper prefers statistical models over machine learning models for three reasons. First, statistical models better align with the purpose of this paper which is to provide traders with a trader friendly model that accurately predicts Bitcoin’s high and low prices so that they are able to make more informed decisions for improved risk management. This is because statistical models are less complex to implement and interpret relative to machine learning models since they do not require advanced technical skills such as coding. This is also because since statistical models are less complex relative to machine learning models, by using statistical models over machine learning models, this ultimately reduces the probability of implementation and interpretation errors (total error) made by cryptocurrency traders who are predominately retail traders without advanced technical skills. Such a reduction in total error coupled with forecast accuracy can enhance risk management of traders which is the original intention of this paper. Second, while machine learning models offer higher predictive power for high frequency data – which is more difficult to manage and therefore not a trader friendly frequency – statistical models offer higher predictive power for low frequency data – which is less difficult to manage and therefore a more trader friendly frequency that better aligns with the needs of the majority of cryptocurrency traders and thus better aligns with the purpose of this paper. For example, in their 2019 paper, Chen, Li and Sun found that after comparing the accuracy rate of Linear Regression (statistical model) and Long Short-Term Memory or LSTM (machine learning model) for predicting Bitcoin’s 5-minute spot price and daily spot price, they found that the machine learning model better predicts the higher of the two frequencies (Bitcoin’s 5-minute spot price), whereas the statistical model better predicts the lower of the two frequencies (Bitcoin’s daily spot price). Similarly, in their 2022 paper, Daniel, Olorunfemi and Oluwatobi found that after comparing forecast evaluations of Linear Regression (statistical model) and Naïve model (machine learning model) for predicting Bitcoin’s daily, weekly, and monthly spot price, they found that the machine learning model better predicts the higher of the three frequencies (Bitcoin’s daily spot price), whereas the statistical model better predicts the lower of the three frequencies (Bitcoin’s weekly and monthly spot price). Third, although machine learning models dominate the literature, statistical models are underexplored, presenting an opportunity to fill this gap by providing a simpler, more accessible solution that reduces total error.

This paper contributes to the literature in five key ways. First, it focuses on Bitcoin’s high and low prices, which are less commonly studied than the spot price. Second, it prioritizes trader friendliness by selecting statistical models and low frequency over more complex machine learning models and high frequency. Third, it conducts model selection by determining the optimal statistical model-frequency combination for predicting Bitcoin’s high and low prices. Fourth, it incorporates Autoregressive Distributed Lags (ARDL) model, which is underexplored within the literature. Finally, it provides traders with a tool to make more informed decisions for improved risk management across varying market conditions, outperforming existing models in the literature. This is because in over 10 years of out of sample testing it achieves an average MAPE of under the benchmark of 10% set by Lewis (1982) in the Journal of Forecasting which states that a MAPE of lower than 10% indicates high forecasting accuracy and higher than 10% indicates low forecasting accuracy. This is also because on average the literature achieves a higher MAPE relative to this paper and the literature also undergoes less significant out of sample testing (see Table V in section III. Results and Discussion, subsection E).

Before delving into answering the research question, it is imperative to establish a foundational understanding of Bitcoin. Introduced by pseudonymous creator Satoshi Nakamoto on January 3rd 2009, Bitcoin was conceived to address various shortcomings within traditional finance. Its primary objectives include providing a more efficient medium of exchange compared to traditional financial institutions and serving as a superior store of value relative to fiat currency. Bitcoin achieves its objectives through features such as its transaction settlement time averaging 10 minutes, facilitated by the Bitcoin mining process, and its deliberately capped supply of 21 million coins.

Furthermore, it is also essential to discuss why Bitcoin holds significant importance for traders. However, before embarking on this discussion, we must first discuss its growth and the rationalities behind such growth. Since its inception, Bitcoin has experienced extraordinary growth, yielding returns exceeding 100,000,000% in less than two decades. This growth surpasses the combined returns of major U.S stock indexes over the past four decades. Three major factors contribute to Bitcoin’s growth. The first factor is Bitcoin’s acceptance as both a medium of exchange and a store of value. Such acceptance is evident in the proliferation of Bitcoin ATMs that have grown by over 6,000% between 2016 and 2022 (Flynn, 2022) and the Security Exchange Commissions (SEC) approval of Bitcoin Spot Exchange Traded Funds (ETF) (Pereira, 2024). The second factor is Bitcoin’s economics, characterized by its increasing marginal cost for miners (S. Hayes, 2019) and diminishing block rewards every four years. The third factor is speculation, as observed in a study conducted by Cai, Chevapatrakul, and V. (2021) who concluded that trader’s propensity to buy or sell is based on recent price movements. Therefore, given Bitcoin’s substantial returns in less than two decades compared to traditional financial assets, due to its acceptance and economics, Bitcoin holds significant importance for traders because it serves as an opportunity to generate significant returns within a short period of time relative to investing in traditional financial assets.

To address the research question, this paper will follow the structured format of: I. Literature Review, II. Methodology, III. Results and Discussion, and IV. Conclusion. Through this approach, this paper aims to contribute valuable insights to the realm of Bitcoin price forecasting.

¹Introduction was edited using ChatGPT for clarity, coherence, conciseness, and structure.

I. Literature Review

A. Section Overview

Since Bitcoin's inception into the world of non-traditional finance, numerous studies have been conducted to analyze Bitcoin's price either for price determinant purposes or price prediction purposes. The difference between the two is that the former focuses on using independent variables to determine which ones drive Bitcoin's price. The latter focuses on using models to determine which model best forecasts the future price of Bitcoin. This section will present the literature related to Bitcoin price analysis by beginning with studies conducted to determine the determinants of Bitcoin's price and ending with studies conducted to predict Bitcoin's price. This section will then discuss the limitations of the literature reviewed and how this relates to this paper's contribution to the literature.

B. Bitcoin Price Determinant Studies

In the realm of Bitcoin price determinant research, there have been several studies conducted to determine which factors drive Bitcoin's price. The findings from this research domain have presented several factors that determine Bitcoin's price that should therefore be incorporated into Bitcoin price prediction models. These include endogenous factors: Bitcoin's hash rate (Bouoiyour and Selmi, 2015) and its mining difficulty (S. Hayes, 2019) and include exogenous factors: Twitter tweet volume and Google search volume for the word "Bitcoin" (Abraham, Ibarra, Higdon and Nelson, 2018; Li and Wang, 2017; Bouoiyour and Selmi, 2015); commodity prices (Lin and An, 2021); U.S macroeconomic indicators (Li and Wang, 2017); U.S financial markets (Ahmed, Bany-Ariffin, Mahat and Nazrul Hisyam, 2020); and herd mentality (Cai, Chevapatrakul and V., 2021). Each of these studies have pioneered the study of Bitcoin's price analysis and have contributed to the research domain of Bitcoin price prediction that will be presented in the next subsection.

C. Bitcoin Price Prediction Studies Using Machine Learning Models

Building upon the insights gained from Bitcoin price determinants studies, researchers have increasingly turned to machine learning models to forecast Bitcoin's price. This subsection delves into the key works of this domain, examining their research question, methodologies, and findings.

Some of the most recent and prominent research papers within the literature of Bitcoin price forecasting using machine learning models includes: Im, Ji and Kim (2019), Alazab, Jay, Kalariya, Kumar, Parmar and Tanwar (2020) and Jiang (2020).

In their 2019 paper, authors Im, Ji and Kim posed the following research question (paraphrased) – which machine learning model best predicts Bitcoin's daily spot price between Deep Neural Networks (DNN), Recurrent Neural Networks (RNN), Convolutional Neural Networks (CNN), Resnet, Long Short-Term Memory (LSTM), and Convolutional Recurrent Neural Networks (CRNN)? To answer this research question, they trained each of these models on a training set spanning from November 2011 to July 2017 using a series of independent variables that were concluded to be Bitcoin price determinants by Bitcoin price determinant studies. These include independent variables such as Bitcoin's block size, mining difficulty, hash rate, market capitalization, miners' revenue, number of transactions, transaction fees, and others. They found LSTM and DNN to be the best predictors of Bitcoin's daily spot price, yielding a

mean absolute percentage error (MAPE) of 4.46% and 4.81%, after testing under two market conditions.

In their 2020 paper, authors Alazab, Jay, Kalariya, Kumar, Parmar and Tanwar posed a different research question relative to Im, Ji and Kim (2019). They asked (paraphrased) – which machine learning model best predicts Bitcoin’s daily spot price between stochastic Multilayer Perception (MLP), stochastic LSTM, deterministic MLP, and deterministic LSTM? To answer this research question, they trained each of these models on a training set spanning from 2017 to 2019 and used similar independent variables as used in Im, Ji and Kim (2019). They found that the stochastic LSTM outperforms the other models within the research question when predicting Bitcoin’s daily spot price, yielding an MAPE of 3.91%, after testing under one market condition.

In his 2020 paper, author Jiang also posed a different research question, this time relative to the previous two studies mentioned. He asked (paraphrased) - which machine learning model best predicts Bitcoin’s hourly spot price between MLP, LSTM, and Gated Recurrent Unit (GRU)? To answer this research question, he trained each of these models on a training set spanning from January 2012 to November 2017 and used Bitcoin’s price lagged 24 hours as the independent variable. He found that the GRU best predicts Bitcoin’s hourly spot price, yielding a root mean squared error (RMSE) of 19.02 in dollar units.

In summary, machine learning models, particularly LSTM and GRU, have shown significant promise in forecasting Bitcoin’s price with relatively low error rates.

D. Bitcoin Price Prediction Studies Using Statistical Models

Now that several studies on Bitcoin price prediction using solely machine learning models have been examined, this subsection will examine research focusing on alternative approaches in this research domain. Specifically, this subsection will focus on studies using only statistical models to predict Bitcoin’s price, presenting their research question, methodologies, and findings.

Some of the most recent and prominent research papers within the literature of Bitcoin price forecasting using statistical models includes: Chilamkurti, M and V (2020), Abu and Rosbi (2017) and Kuen Siu (2022).

In their 2020 paper, authors Chilamkurti, M and V posed the following research question (paraphrased) – can the Autoregressive Integrated Moving Average (ARIMA) predict Bitcoin’s daily closing price and Bitcoin’s hourly closing price? To answer this research question, they trained the ARIMA model on a training set spanning from April 2013 to December 2016 and used the lags of Bitcoin’s daily closing price as the independent variable for forecasting Bitcoin’s daily closing price and used lags of Bitcoin’s hourly closing price as the independent variable for forecasting Bitcoin’s hourly closing price. They found that the ARIMA model can predict Bitcoin’s daily closing price, achieving an accuracy rate of 49%. However, the ARIMA model cannot predict Bitcoin’s hourly closing price.

In their 2017 paper, authors Abu and Rosbi posed a different research question relative to Chilamkurti, M and V. They asked (paraphrased) – can ARIMA predict Bitcoin’s monthly spot price? To answer this research question, they trained the ARIMA model on a training set spanning from January 2013 to August 2017 using two lags of Bitcoin’s monthly spot price and two lags of the moving average as the independent variables. They found that the ARIMA model can predict Bitcoin’s monthly spot price with high accuracy with a MAPE of 5.36%, after testing under one market condition.

In his 2022 paper, author Kuen Siu also posed a different research question, this time relative to the previous two studies discussed. He asked (paraphrased) – which statistical model between Autoregressive Self Exciting Threshold (AR-Setar), Generalized Autoregressive Conditional Heteroskedasticity (GARCH), and Stochastic Volatility (SV), best predict Bitcoin's daily closing price under different states of volatility? To answer this research question, he trained each of these models on a training set spanning from January 2018 to December 2020 using the lag of Bitcoin's daily closing price as the independent variable. He found that the SV model best predicts Bitcoin's daily closing price under different states of volatility relative to the AR-Setar and GARCH models, achieving a log likelihood of approximately -3443.83.

In summary, statistical models, specifically ARIMA and SV, have also shown significant promise in accurately forecasting Bitcoin's price.

E. Bitcoin Price Prediction Studies Using Both Machine Learning Models and Statistical Models

Now that multiple studies on Bitcoin price prediction using exclusively machine learning models or statistical models have been examined, this subsection will now focus on studies that employ both machine learning models and statistical models to predict Bitcoin's price. These studies will be presented by discussing their research questions, methodologies, and findings.

Some of the most recent and prominent research papers within the literature of Bitcoin price forecasting using both machine learning models and statistical models includes: Chen, Li and Sun (2019), Ho, Kumar and Vatambeti (2021), Daniel, Olorunfemi and Oluwatobi (2022), Alon, Hassan Shakil and Haque Munim (2019) and Tripathi and Tripathi (2022).

In their 2019 paper, Chen, Li and Sun posed the following research question (paraphrased) – which model, between statistical models Logistic Regression and Linear Discriminant Analysis, and machine learning models Random Forest, XGBoost, Quadratic Discriminant Analysis, Support Vector Machine, and LSTM, best predicts Bitcoin's daily spot price and best predicts Bitcoin's 5-minute spot price? To answer this research question, they trained each of these models on a training set spanning from February 2017 to February 2019 using similar independent variables that were used in Im, Ji and Kim (2019), to determine which best predicts Bitcoin's daily spot price. They also trained each of these models on a training set spanning from July 2017 to January 2018, also using similar independent variables as used in Im, Ji and Kim (2019), to determine which best predicts Bitcoin's 5-minute spot price. They found that the statistical model Logistic Regression best predicts Bitcoin's daily spot price with an accuracy rate of 66%, whereas the machine learning model LSTM best predicts Bitcoin's 5-minute spot price with an accuracy rate of 67.2%.

In their 2021 paper, Ho, Kumar and Vatambeti posed a different research question from Chen, Li and Sun. They asked – which model between statistical model Linear Regression and machine learning model LSTM, best predicts Bitcoin's daily spot price? To answer this research question, they trained each of these models on a training set spanning from August 2017 to August 2020 using Bitcoin's daily close price, open price, high price, and low price as independent variables. They found that the machine learning model LSTM best predicts Bitcoin's daily spot price relative to the statistical model Linear Regression.

In their 2022 paper, Daniel, Olorunfemi and Oluwatobi also posed a different research question, this time relative to the previous two studies mentioned. They asked – which model between statistical models Linear Regression and ARIMA and machine learning models Naïve, Exponential Smoothing, Neural Network, STL, and Holt Winters Filer, best predicts Bitcoin's

daily, weekly, and monthly spot price? To answer this research question, they trained each of these models on a training set spanning from January 2015 to November 2021 using the lags of the dependent variable as the independent variables. They found that the machine learning model Naïve best predicts Bitcoin's daily spot price while the statistical model Linear Regression best predicts Bitcoin's weekly and monthly spot price.

In their 2019 paper, Alon, Hassan Shakil and Haque Munim also posed a different research question, this time relative to the previous three studies discussed. They asked – which model between statistical model ARIMA and machine learning model Neural Network Autoregressive (NNAR), best predicts Bitcoin's daily spot price? To answer this research, they trained each of these models on a training set spanning from January 2012 to June 2017 using lags of Bitcoin's daily spot price as the independent variable. They found that the ARIMA model best predicts Bitcoin's daily spot price relative to NNAR, yielding an MAPE of .37%, after testing under four market conditions.

In their 2022 paper, Tripathi and Tripathi also posed a different research question, this time relative to the previous four studies mentioned. They asked – which model between statistical model ARIMA and machine learning models Random Walk (RW), Artificial Neural Network (ANN), and the Ensemble Model best predicts Bitcoin's daily spot price? To answer this research question, they trained each of these models on a training set spanning from April 2013 to June 2020 using lags of Bitcoin's daily spot price as the independent variable. They found that the Ensemble Model best predicts Bitcoin's daily spot price relative to the statistical model ARIMA and the machine learning models RW and ANN.

In summary, studies that employ machine learning models and statistical models show varying results depending on the frequency of prediction. While machine learning models outperform statistical models at high frequency forecasting, particularly short-term price movements, statistical models outperform machine learning models at low frequency forecasting, particularly long-term price movements.

²For a table representation of the literature reviewed in subsections C to E of the Literature Review, refer to Appendix I, II, and III.

F. Contributions to the Literature

From the literature reviewed within the scope of Bitcoin price prediction, several common themes exist amongst these studies. First, most of these studies forecast Bitcoin's spot price, whereas forecasting Bitcoin's high price and low price is underexplored. Because forecasting the latter can enhance risk management by enabling traders to set precise entry and exit points, most of these studies fail to enhance risk management. Second, most of these studies employ machine learning models in lieu of statistical models to forecast Bitcoin's price. This is evident in the nineteen machine learning models, relative to the seven statistical models, discussed in subsections C to E of the Literature Review. Because machine learning models are more complex than statistical models, most of these studies fail to provide the majority of cryptocurrency traders – retail traders with basic to intermediate forecast skills – with a trader friendly model that they can use without an increased probability of implementation and interpretation error that may exceed the total error – implementation, interpretation, and prediction error – of statistical models. Third, most of the literature fails to optimize their predictions by conducting model selection to determine the optimal model frequency combination for forecasting Bitcoin's price. Fourth, although the literature incorporates statistical models for predicting Bitcoin's price, the use of Autoregressive Distributed Lags (ARDL) models is underexplored. Fifth, despite high forecasting accuracy, several of the literature's models have undergone limited robustness testing, with four market conditions being the maximum, indicating that these models are limited in their adaptability and generalizability to various market conditions. Clearly, each of these common themes represent major gaps within the literature that need to be filled in order to enhance traders risk management.

As a contribution to the literature, this paper will enhance traders risk management by bridging the gap within the literature. This will be done in several ways. First, this paper will forecast Bitcoin's high price and low price. Second, this paper will provide traders with a trader friendly model by selecting statistical models over machine learning models. For the sake of trader friendliness and statistical models' superior accuracy at forecasting low frequency, as discussed in subsection E of the Literature Review, this paper will focus on low frequency in lieu of high frequency. Third, this paper will optimize forecast by conducting model selection to determine the optimal statistical model frequency combination that best predicts – in terms of MAPE, AIC, and SIC – Bitcoin's high price and low price. Fourth, this paper will incorporate Autoregressive Distributed Lags (ARDL) models. Fifth, this paper will conduct significant and comprehensive robustness testing encompassing over six market conditions and Bitcoin's full historical life, to ensure that the statistical model frequency combination selected is both adaptable and generalizable across various market conditions and can therefore be used by traders for enhanced risk management despite volatility and market shifts.

II. Methodology

A. Methodology Benchmark Papers

This paper follows the methodology employed by two notable academic papers published in reputable journals – The Journal of Economics and Finance and The Journal of Risk and Financial Management.

The first methodology benchmark paper titled, *Empirical Analysis of Bitcoin Price*, authored by Yuanyuan Chen, employs a series of advanced statistical models to answer its specified research question. The second methodology benchmark paper titled, *A Gated Recurrent Unit Approach to Bitcoin Price Prediction*, authored by Meheli Basu, Aniruddha Dutta, and Saket Kumar, employs advanced machine learning models to answer its specified research question. This paper follows the methodology of the first and second methodology benchmark papers for two reasons. First, the independent variables employed by both methodology benchmark papers are vital to answering the research question of this paper. This is because both methodology benchmark papers have proven that these independent variables are major predictors of Bitcoin's price. Second, the robustness tests employed by the first methodology benchmark paper are also vital to answering the research question of this paper. This is because such robustness tests factors in multiple market conditions that are vital for examining this paper's models performance across various market environments.

This paper will utilize similar methodologies as employed by the first and second methodology benchmark papers but with minor differences that are tailored to address this paper's research question and objectives. First, like the first and second methodology benchmark papers, this paper will utilize the same independent variables deemed to be determinants of Bitcoin's price. These include endogenous factors or variables that are internal to the Bitcoin network such as Bitcoin's market capitalization, transaction volume, hash rate, price volatility, miner's revenue, transaction fees, circulating supply, and block size. These independent variables also include exogenous factors or variables that are external to the Bitcoin network such as WTI Crude Oil Spot Price, Dow Jones Index, Google Trends, U.S Federal Funds Rate, Gold Spot Price, USD Index, S&P500 Index, 10 Year U.S Bond Yield. Second, dissimilar to the first and second methodology benchmark papers who applied models such as a Vector Error Correction (first methodology benchmark paper) and models such as the advanced Gated Recurrent Unit (second methodology benchmark paper), in this paper these independent variables will be applied to a standard regression model known as Ordinary Least Squares (OLS). There are two reasons for applying OLS. One reason is that it is the easiest statistical model to implement and interpret, thus best meeting the second half of this paper's objective. Another reason is that, as OLS aligns with this objective, it will serve as a baseline regression to be compared with a non-standard regression that is generally less comprehensible. This comparison aims to determine which statistical model between the two better addresses the research question. Third, like the first methodology benchmark paper, because the first methodology benchmark paper highlighted the statistical significance of incorporating lagged values of Bitcoin's price, this paper will employ a lagged time series model as the non-standard regression model. The non-standard regression model that will be employed within this paper is the Autoregressive Distributed Lags (ARDL) model. This is because compared to all lagged time series models such as the first methodology benchmark papers Vector Error Correction Model (VECM), the ARDL model is one of the easiest to implement and interpret. Therefore, it stands out as one of the best options, among lagged time series models, for meeting this paper's secondary objective. This is also because research shows

that while ARDL modeling has been underexplored within the literature on Bitcoin price forecasting, it has been explored when forecasting other variables and for these variables it outperforms other models. For example, in their 2022 paper titled, *Gold Price Forecasting Using Multivariate Stochastic Model*, authors Chatterjee, Madziwa, and Pillalamarry compared ARDL, Autoregressive Integrated Moving Average (ARIMA), and Stochastic Gaussian to determine which best predicts the price of Gold. They found the best performing model to be ARDL. Similarly, in their 2017 paper titled, *Palm Oil Forecasting Model: An Autoregressive Distributed Lag (ARDL) Approach*, authors Hamid and Shabri also compared ARDL and ARIMA. However, in their study they aimed to determine which best predicts the price of Palm Oil. They found the best performing model to also be ARDL. Therefore, incorporating ARDL into this paper serves as an excellent opportunity to determine if the ARDL model can also outperform other models – particularly OLS – when forecasting Bitcoin’s high and low price, similar to its behavior when forecasting other variables. Fourth, like the first methodology benchmark paper, this paper will test the robustness of the ARDL model under the same market regimes that the first methodology benchmark paper tested its VECM: bull market and bear market. This will provide a comprehensive approach to model robustness.

Overall, both the first and second methodology benchmark papers will be utilized within this paper as methodological blueprints for answering the research question.

B. Data

This paper includes 23 independent variables that have each been categorized as an endogenous variable or exogenous variable. These independent variables have been selected to be included in this paper because both the literature review and the methodology benchmark papers conclude that these independent variables are predictors of Bitcoin’s price. The endogenous variables represent predictors of Bitcoin’s price that are internal to Bitcoin since they derive from the Bitcoin Network. They include Bitcoin’s open price, stripped block size, price volatility, mining difficulty, hash rate, market capitalization, miner’s revenue, number of transactions, circulating supply, and total transaction fees. The exogenous variables represent predictors of Bitcoin’s price that are external to Bitcoin since they are either related to the U.S macroeconomy, U.S financial markets, or Bitcoin sentiment. They include the U.S Federal Funds Rate, U.S GDP, U.S CPI Inflation Rate, U.S M2 Money Supply, Dow Jones Index Open Price, Gold Open Price, Gold Open Volatility, Oil Open Price, Oil Open Volatility, S&P500 Index Open Price, U.S 10 Year Bond Open Yield, USD Index Open Price, and Google Trend. This paper also includes 2 dependent variables. They include Bitcoin’s high price and low price.

The data of the independent variables was extracted from several sources. The data for each of the endogenous variables was extracted from Bitcoinvisuals.com except for Bitcoin’s open price which was extracted from Investing.com and Bitcoin’s price volatility which was extracted from Data.Bitcoinity.com. The data for the exogenous variables was extracted from The Federal Reserve Bank of St. Louis (FRED), Investing.com, and Google Search Engine. The data of the dependent variables was also extracted from Investing.com. The variable definitions and data sources can be found in Appendix IV, V, and VI.

Each of the independent variables and dependent variables were extracted from their specified source for the time period of October 2011 to October 2023 and at the daily frequency. This time period represents one of the earliest dates and one of the latest dates that dependent variable data (Bitcoin high price and low price) was available on Investing.com. For researchers

replicating this study using a different asset, it is recommended to extract data for the independent and dependent variables using the earliest and latest date that data for the dependent variable is available to ensure a comprehensive analysis. Note that the exogenous variables that were extracted from FRED and Google Search Engine were extracted for the specified time period but at a lower frequency. Because of this, the data of these variables were the first to be extracted and imported into SAS where they were converted into the daily frequency.

Upon data extraction of the remaining independent variables and dependent variables, the data for each were imported into SAS where they underwent data processing. This entailed merging the datasets of each of these variables together and then cleaning the merged dataset for data quality by removing missing values and duplicates. Once the merged dataset had been cleaned, its final range spanned from October 2011 to October 2020 with 1,941 daily observations.

This original merged dataset was then used to create two new merged datasets. The first dataset created was the merged dataset converted into weekly frequency and the second dataset created was the merged dataset converted into monthly frequency. The final number of merged datasets created were three – one for the daily frequency, one for the weekly frequency, and one for the monthly frequency. The two new merged datasets were cleaned for data quality by removing missing values and duplicates. Once the two new merged datasets had been cleaned, their final range spanned from October 2011 to October 2020 with 338 weekly observations and December 2011 to October 2020 with 85 monthly observations.

The three merged datasets will be used to construct the standard regression (OLS), whereas the original Bitcoin price dataset, extracted from Investing.com for the time period of October 2011 to October 2023 at the daily frequency, will be used to construct the non-standard regression (ARDL). This was done to increase the probability of the ARDL model generating high forecasting accuracy by training it on more observations that were truncated during the merger. The original Bitcoin price dataset was cleaned using SAS, removing missing values and duplicates. The final dataset spanned from October 2012 to October 2023 and consisted of 4,018 daily observations. This dataset was then used to create two additional new Bitcoin price datasets making a total of three Bitcoin price datasets— one for the daily frequency, one for the weekly frequency, and one for the monthly frequency. These new Bitcoin price datasets were also cleaned for data quality by removing missing values and duplicates. Once the two new Bitcoin price datasets had been cleaned, their final range spanned from October 2012 to October 2023 with 574 weekly observations and October 2012 to October 2023 with 133 monthly observations.

C. The Models

C.1. Subsection Overview

This subsection will develop two types of statistical models using EViews. These models include a standard regression (OLS) and a non-standard regression (ARDL). The purpose of developing these two types of models is to determine which of the two better answers the research question – the model that, of all the statistical models, best meets the secondary objective of this paper (OLS) or the model that, of all the statistical models, does not best meet the secondary objective but incorporates a major determinant of Bitcoin's price according to the first methodology benchmark paper (the lagged value of Bitcoin's price) and is among the best lagged time series models to meet the secondary objective of this paper (ARDL).

The ARDL model will be developed using regime dependent analysis. This will result in three types of ARDL models – General ARDL model, ARDL Bull Market model, and ARDL Bear Market model. There are two major reasons for developing the ARDL model using regime dependent analysis. First, the ARDL models trained under different regimes yield different forecasting results due to differences in data structures caused by differences in regimes. Second, due to these differences in forecasting results, encompassing ARDL models trained under different regimes is crucial for a comprehensive and due diligent approach to answering the research question by comparing these ARDL models to each other and to the OLS model.

For Bitcoin’s high price and for Bitcoin’s low price, each of these statistical models (OLS, General ARDL, ARDL Bull, and ARDL Bear) will be developed across multiple frequencies (daily, weekly, and monthly) to determine which statistical model frequency combination best predicts Bitcoin’s high price and Bitcoin’s low price or in other words, which statistical model frequency combination best answers the research question.

C.2. OLS Model Development

For Bitcoin’s high price and Bitcoin’s low price, the OLS model was developed across multiple frequencies (daily, weekly, and monthly) and was trained on the three merged datasets ranging from the period of October 2011 to October 2020. This resulted in a total of three OLS models for Bitcoin’s high price (one for each frequency) and three OLS models for Bitcoin’s low price (one for each frequency). Each OLS model encompassed the log transformation of the 23 independent variables regressed against the log transformation of the dependent variables (Bitcoin’s high price and Bitcoin’s low price). These variables were log transformed to enable the OLS model to have a better view of the trends within the data so that the model can better learn these trends and therefore yield more accurate forecast of Bitcoin’s high price and Bitcoin’s low price. For a table representation of the OLS models developed, refer to Appendix VII.

C.3. General ARDL Model Development

For Bitcoin’s high price and Bitcoin’s low price, the General ARDL was also developed across multiple frequencies (daily, weekly, and monthly) but was trained on the three Bitcoin price datasets ranging from the period of October 2012 to October 2023. This resulted in a total of three General ARDL models for Bitcoin’s high price (one for each frequency) and three General ARDL models for Bitcoin’s low price (one for each frequency).

The purpose of developing the General ARDL model was to determine how well this model forecast Bitcoin’s high price and Bitcoin’s low price when trained under general market conditions and to serve as a benchmark against both the OLS model and the ARDL Bull Market and ARDL Bear Market models to determine which best answers the research question.

To develop each General ARDL model for Bitcoin’s high price and Bitcoin’s low price, several steps were executed. The next four paragraphs discuss those steps in detail.

First, each of the 23 independent variables as well as each of the dependent variables (Bitcoin’s high price and Bitcoin’s low price) were log transformed. This was done for the same reasons that the log transformations were used in the OLS model as previously discussed in sub-subsection C.2.

Second, each of the 23 independent variables were individually regressed against the dependent variables. This was done to determine their significance at forecasting Bitcoin’s high price and Bitcoin’s low price based on their mean absolute percent error (MAPE) – a measure of

the average absolute percentage error between the independent variables predicted values of Bitcoin's high price and Bitcoin's low price and the actual value of Bitcoin's high price and Bitcoin's low price. The MAPE of each independent variable was examined. Those with a MAPE of greater than 10% were omitted from the General ARDL model while those with a MAPE of less than 10% were considered. This MAPE criteria was selected based off Lewis (1982) who argues that a MAPE of less than 10% indicates high forecast accuracy and anything above 10% indicates low forecast accuracy. Ultimately, the independent variable with the lowest MAPE was selected to be the sole independent variable included within the General ARDL. This ensures that only the most significant predictor of Bitcoin's high price and Bitcoin's low price was included, thereby simplifying the model, and better aligning it with the secondary objective of this paper. The independent variable chosen as being the most significant predictor of Bitcoin's high price and Bitcoin's low price was Bitcoin's open price.

Third, an Augmented Dickey Fuller (ADF) stationarity test of Bitcoin's high price, Bitcoin's low price, and Bitcoin's open price was conducted at the 5% significance level. The test for stationarity was executed in order to conduct optimal General ARDL model specification.

Fourth, after observing that each variable is stationary in the first difference at the 5% significance level, with a P-value of 0.00, a correlogram of each variable transformed into the first difference was examined. The purpose of examining a correlogram of Bitcoin's high price and Bitcoin's low price in the first difference was to determine the optimal lags of Bitcoin's high price and Bitcoin's low price that should be incorporated into the General ARDL in order to yield optimal forecasting results of Bitcoin's high price and Bitcoin's low price. Similarly, the purpose of examining a correlogram of Bitcoin's open price in the first difference was to determine the optimal lags of Bitcoin's open price that should be incorporated into the General ARDL in order to yield optimal forecasting results for Bitcoin's high price and Bitcoin's low price.

Based on the correlograms partial correlation function, the optimal lags for Bitcoin's high price and Bitcoin's open price and thus the optimal General ARDL model for predicting Bitcoin's high price at the daily, weekly, and monthly frequency are General ARDL (1,8), (2,15), and (1,2).

On the contrary, based on the correlograms partial correlation function, the optimal lags for Bitcoin's low price and Bitcoin's open price and thus the optimal General ARDL for predicting Bitcoin's low price at the daily, weekly, and monthly frequency are General ARDL (6,5), (2,15), and (5,2). For a table representation of the General ARDL models developed, refer to Appendix VIII.

C.4. ARDL Bull and Bear Market Model Development

For Bitcoin's high price and Bitcoin's low price, the ARDL Bull Market model was also developed across multiple frequencies (daily, weekly, and monthly) and its lag structure was selected using the three Bitcoin price datasets full range of October 2012 to October 2023. However, this model was not trained on the full dataset range. Instead, it was trained on the most recent major cryptocurrency bull market – the 2020 bull run. This ranged from the period of March 2020 to November 2021. This resulted in three ARDL Bull Market models for Bitcoin's high price (one for each frequency) and three ARDL Bull Market models for Bitcoin's low price (one for each frequency). For a table representation of the ARDL Bull Market models developed, refer to Appendix IX.

There are several reasons why the ARDL Bull Market model was trained on the most recent cryptocurrency bull market while using the three Bitcoin price datasets full range of

October 2012 to October 2023 to select its lag structure. First, the ARDL Bull Market model was trained on a cryptocurrency bull run to see how it would perform when trained under this regime. Second, the ARDL Bull Market model was trained on the most recent cryptocurrency bull run so that the model learns the latest market trends within the data so that it can more accurately forecast Bitcoin's future high and low price. Third, the ARDL Bull Market models lag structure was selected using the three Bitcoin price datasets full range of October 2012 to October 2023 which reflect the bulk of Bitcoin's life, to ensure that the ARDL Bull Market model doesn't solely perform well on bull runs but can also generalize to other market conditions when tested.

The purpose of developing the ARDL Bull Market model is for the model to serve as a benchmark against both the OLS model and the General ARDL and ARDL Bear Market models to determine which best answers the research question.

In addition, for Bitcoin's high price and Bitcoin's low price, the ARDL Bear Market model was also developed across multiple frequencies (daily, weekly, and monthly) and its lag structure was also selected using the three Bitcoin price datasets full range of October 2012 to October 2023. However, similar to the ARDL Bull Market model, it was not trained on the full dataset range. Instead, it was trained on one of the most recent major cryptocurrency bear markets – the 2017 bear run. This ranged from December 2017 to March 2020. This resulted in three ARDL Bear Market models for Bitcoin's high price (one for each frequency) and three ARDL Bear Market models for Bitcoin's low price (one for each frequency). For a table representation of the ARDL Bear Market models developed, refer to Appendix X.

The ARDL Bear Market model was trained on one of the most recent major cryptocurrency bear markets while using the three Bitcoin price datasets full range of October 2012 to October 2023 to select its lag structure for the same reasons the ARDL Bull Market model was trained on the most recent cryptocurrency bull market while using the three Bitcoin price datasets full range to select its lag structure.

The purpose of developing the ARDL Bear Market model is for the model to serve as a benchmark against both the OLS model and the General ARDL and ARDL Bull Market models to determine which best answers the research question.

C.5. Conclusion

Overall, approximately 24 statistical models were created across multiple frequencies. This was done to allow for a comprehensive analysis for determining the best statistical model frequency combination that best predicts Bitcoin's high price and Bitcoin's low price.

D. Robustness Tests

D.1. Subsection Overview

This subsection will firstly discuss the comparative analysis employed to determine the best statistical model per frequency (daily, weekly, and monthly) for predicting Bitcoin's high price and Bitcoin's low price. Once the former is determined, this subsection will secondly discuss the comparative analysis employed to determine the best statistical model frequency combination for predicting Bitcoin's high price and Bitcoin's low price. Lastly, once the research question is answered, this subsection will discuss the robustness tests conducted of the statistical model frequency combination that best answers the research question.

D.2. Comparative Analysis for the Best Statistical Model Per Frequency for Predicting Bitcoin's High Price and Bitcoin's Low Price

To determine the best statistical model per frequency – daily, weekly, and monthly – for predicting Bitcoin's high price and Bitcoin's low price, each model within a given frequency was compared using their training set mean absolute percent error (MAPE) as well as Akaike Information Criterion (AIC) and Schwarz Information Criterion (SIC). This performance metric and criteria are the most commonly used in time series analysis, and for this reason, were selected over other performance metrics and criteria. As previously stated, the MAPE measures the average absolute percentage error between the model's predicted value and actual value. In other words, it measures how accurate the model is at predicting Bitcoin's high price and Bitcoin's low price. The AIC and SIC measure the goodness of fit of the model while factoring in model complexity. In other words, it measures how well the models' predicted values fit the actual values while penalizing for model complexity. Therefore, the model with the lowest MAPE and lowest AIC and SIC within a given frequency was selected as the best for that frequency for predicting Bitcoin's high price and Bitcoin's low price. For a table representation of the statistical models compared per frequency for predicting Bitcoin's high price and Bitcoin's low price, refer to Appendix XI and XII.

D.3. Comparative Analysis for the Best Statistical Model Frequency Combination for Predicting Bitcoin's High Price and Bitcoin's Low Price

To determine the best statistical model frequency combination for predicting Bitcoin's high price and Bitcoin's low price, the best statistical model selected per frequency – daily, weekly, and monthly – were each compared to one another in terms of training set MAPE, AIC, and SIC. The statistical model frequency combination with the lowest of this performance metric and criteria was concluded to be the best statistical model frequency combination for predicting Bitcoin's high price and Bitcoin's low price.

D.4. Robustness Tests for the Best Statistical Model Frequency Combination for Predicting Bitcoin's High Price and Bitcoin's Low Price

After determining the best statistical model frequency combination for predicting Bitcoin's high price and Bitcoin's low price, robustness tests were executed using Excel.

The purpose of executing robustness tests was to determine whether the selected statistical model frequency combination has the potential to be used in the real world for risk management, where it doesn't perform well solely on seen data (training set) but also performs well on unseen data (out of sample test) and across multiple market conditions.

In total there were 14 robustness tests performed ranging from 2011 to 2024 and using investing.com data. Of these 14 robustness tests, seven were for Bitcoin's high price and seven were for Bitcoin's low price. These seven included five out of sample back test that encompassed the 2012 bull run, 2013 bear run, 2016 bull run, 2017 bear run, and Bitcoin's full historical life. These seven also included two out of sample forward tests that encompassed the 2021 bear run and 2023 mini bull run. The bull run start dates were selected at the start of the accumulation phase which is a period where Bitcoin's price begins to surge right before and in anticipation of the halving. The halving is an event that occurs every four years where the amount of Bitcoin that Bitcoin miners receive is cut in half in order to maintain a fixed supply of 21 million Bitcoins.

The bull run end dates were selected within the twelve to eighteen months of the halving, which is the typical length of a bull run. The bear run start dates were selected on the end date of the bull run. The bear run end dates were selected on the start date of the next accumulation phase. The specific dates for these tests can be found in section III. Results and Discussion, subsection C and D. These models' robustness was determined based on MAPE for each robustness test.

III. Results and Discussion

A. Training Set Results of Statistical Models Predicting Bitcoin's (BTC) High Price

Table I on page 18 displays the training set results of each statistical model constructed to predict Bitcoin's high price at the daily, weekly, and monthly frequency. The purpose of Table I is to show how each statistical model frequency combination for predicting Bitcoin's high price compares to one another to determine the best (in terms of training set MAPE, AIC, and SIC) among these statistical model frequency combinations for predicting Bitcoin's high price.

Panel A of Table I displays the training set results of the statistical models developed to predict Bitcoin's high price at the daily frequency. From this table, it is evident that each of these statistical models exhibited an exceptional MAPE ranging from 0.22% to 0.41%. This is because such an MAPE is below the Lewis (1982) benchmark of 10% in the Journal of Forecasting as discussed in section II. Methodology, subsection C.3. of this paper. Despite each of these models' high level of accuracy, the statistical model with the best results encompassing the lowest MAPE and the lowest AIC and SIC is the ARDL Bear Market model with a MAPE of 0.22% and AIC and SIC of -4.4 and -4.3, respectively. Therefore, this statistical model was selected as the best for predicting Bitcoin's high price at the daily frequency.

Panel B of Table I displays the training set results of the statistical models constructed to predict Bitcoin's high price at the weekly frequency. Similarly, from this table, it is evident that each of these statistical models also exhibited an exquisite MAPE with a value ranging from 0.15% to 0.46%. This is because such a MAPE is also below the Lewis (1982) benchmark of 10% in the Journal of Forecasting as discussed in section II. Methodology, subsection C.3. of this paper. Although each of these models yielded high levels of accuracy, the statistical model with the lowest MAPE and the lowest AIC and SIC is the ARDL Bull Market model with a MAPE of 0.15% and AIC and SIC of -4.6 and -4.0, respectively. Therefore, this statistical model was selected as the best for predicting Bitcoin's high price at the weekly frequency.

Panel C of Table I displays the training set results of the statistical models developed to predict Bitcoin's high price at the monthly frequency. Likewise, from this table, it is evident that each of these statistical models also demonstrated an excellent MAPE with a value ranging from 0.17% to 0.30%. This is because such a MAPE is also below the Lewis (1982) benchmark of 10% in the Journal of Forecasting as discussed in section II. Methodology, subsection C.3. of this paper. Regardless of each of these models' high level of accuracy, the statistical model with the lowest MAPE and the lowest AIC and SIC is the ARDL Bear Market model with a MAPE of 0.17% and AIC and SIC of -4.4 and -4.2, respectively. Therefore, this statistical model was selected as the best for predicting Bitcoin's high price at the monthly frequency.

Lastly, Panel D of Table I displays the training set results of the best statistical model frequency combination for predicting Bitcoin's high price. After comparing the training set results of each frequency's (daily, weekly, and monthly) best statistical model for predicting Bitcoin's high price, the statistical model frequency combination with the lowest MAPE and the lowest AIC and SIC is the ARDL Bull Market model for the weekly frequency with a MAPE of 0.15%

and AIC and SIC of -4.6 and -4.0, respectively. Therefore, this statistical model frequency combination is the BEST for predicting Bitcoin's high price.

Table I: Training Set Results of Statistical Models Predicting Bitcoin's (BTC) High Price

<i>Panel</i>	<i>Model</i>	<i>MAPE</i>	<i>AIC</i>	<i>SIC</i>
Panel A: <i>Daily Frequency</i>	Baseline: OLS	0.377744	-3.711922	-3.643039
	General ARDL	0.41234	-3.642858	-3.625586
	ARDL Bull	0.2159	-4.335998	-4.256005
	ARDL Bear	0.228421	-4.400413	-4.360018
Panel B: <i>Weekly Frequency</i>	Baseline: OLS	0.466701	-3.630045	-3.358586
	General ARDL	0.283803	-4.244556	-4.097514
	ARDL Bull	0.152175	-4.611675	-4.076796
	ARDL Bear	0.180097	-4.52065	-4.072092
Panel C: <i>Monthly Frequency</i>	Baseline: OLS	0.300507	-4.5062	-3.81651
	General ARDL	0.253504	-4.247376	-4.137636
	ARDL Bull	0.213952	-3.877618	-3.628923
	ARDL Bear	0.171039	-4.490788	-4.252894
Panel D: <i>Best Statistical Model Frequency Combination</i>	ARDL Bear Daily	0.228421	-4.400413	-4.360018
	ARDL Bull Weekly	0.152175	-4.611675	-4.076796
	ARDL Bear Monthly	0.171039	-4.490788	-4.252894

³Table I shows the training set MAPE, AIC, and SIC results of each statistical model developed to predict Bitcoin's high price at the daily, weekly, and monthly frequency. The purpose of Table I is to determine the best statistical model frequency combination for predicting Bitcoin's high price. This is done by determining the best statistical model developed to predict Bitcoin's high price at the daily frequency (Panel A), weekly frequency (Panel B), and monthly frequency (Panel C) and then comparing each of those individual frequencies best statistical model training set MAPE, AIC, and SIC to determine the best statistical model frequency combination for predicting Bitcoin's high price (Panel D). Such comparison in Panel D shows that the best statistical model frequency for predicting Bitcoin's high price is the ARDL Bull Weekly model. This is because it yielded the lowest of the three performance metrics (MAPE, AIC, and SIC).

B. Training Set Results of Statistical Models Predicting Bitcoin's (BTC) Low Price

Table II on page 20 displays the training set results of each statistical model constructed to predict Bitcoin's low price at the daily, weekly, and monthly frequency. The purpose of Table II is to show how each statistical model frequency combination for predicting Bitcoin's low price compares to one another to determine the best (in terms of training set MAPE, AIC, and SIC) among these statistical model frequency combinations for predicting Bitcoin's low price.

Panel A of Table II displays the training set results of the statistical models developed to predict Bitcoin's low price at the daily frequency. From this table, it is evident that each of these statistical models displayed a remarkable MAPE ranging from 0.21% to 0.55%. This is because such an MAPE is below the Lewis (1982) benchmark of 10% in the Journal of Forecasting as discussed in section II. Methodology, subsection C.3. of this paper. Regardless of each of these models' high level of accuracy, the statistical model with the lowest MAPE and lowest AIC and SIC is the ARDL Bull Market model with a MAPE of 0.21% and an AIC and SIC of -4.0 and -3.9, respectively. Therefore, this statistical model was selected as the best for predicting Bitcoin's low price at the daily frequency.

Panel B of Table II displays the training set results of the statistical models constructed to predict Bitcoin's low price at the weekly frequency. Similarly, from this table, it is evident that each of these statistical models also displayed an incredible MAPE ranging from 0.23% to 0.70%. This is because such a MAPE is also below the Lewis (1982) benchmark of 10% in the Journal of Forecasting as discussed in section II. Methodology, subsection C.3. of this paper. Despite each of these models' high level of accuracy, the statistical model with the lowest MAPE and lowest AIC and SIC is the ARDL Bear Market model with a MAPE of 0.23% and an AIC and SIC of -4.1 and -3.6, respectively. Therefore, this statistical model was selected as the best for predicting Bitcoin's low price at the weekly frequency.

Panel C of Table II displays the training set results of the statistical models developed to predict Bitcoin's low price at the monthly frequency. Likewise, from this table, it is evident that each of these statistical models also displayed an exquisite MAPE ranging from 0.13% to 0.51%. This is because such a MAPE is also below the Lewis (1982) benchmark of 10% in the Journal of Forecasting as discussed in section II. Methodology, subsection C.3. of this paper. Although each of these models yielded a high level of accuracy, the statistical model with the lowest MAPE and the lowest AIC and SIC is the ARDL Bull Market model with a MAPE of 0.13% and an AIC and SIC of -4.4 and -3.9, respectively. Therefore, this statistical model was selected as the best for predicting Bitcoin's low price at the monthly frequency.

Panel D of Table II displays the training set results of the best statistical model frequency combination for predicting Bitcoin's low price. After comparing the training set results of each frequency's (daily, weekly, and monthly) best statistical model for predicting Bitcoin's low price, the statistical model frequency combination with the lowest MAPE and the lowest AIC and SIC is the ARDL Bull Market model for the monthly frequency with a MAPE of 0.13% and AIC and SIC of -4.4 and -3.9, respectively. Therefore, this statistical model frequency combination is the BEST for predicting Bitcoin's low price.

Table II: Training Set Results of Statistical Models Predicting Bitcoin's (BTC) Low Price

<i>Panel</i>	<i>Model</i>	<i>MAPE</i>	<i>AIC</i>	<i>SIC</i>
Panel A: Daily Frequency	Baseline: OLS	0.556801	-2.80179	-2.73291
	General ARDL	0.457504	-3.14594	-3.12553
	ARDL Bull	0.21562	-4.04453	-3.95726
	ARDL Bear	0.296662	-3.69855	-3.62353
Panel B: Weekly Frequency	Baseline: OLS	0.703277	-2.72757	-2.45611
	General ARDL	0.318689	-3.69731	-3.55026
	ARDL Bull	0.23026	-3.68376	-3.14888
	ARDL Bear	0.238608	-4.10603	-3.65747
Panel C: Monthly Frequency	Baseline: OLS	0.365627	-4.21365	-3.52396
	General ARDL	0.519908	-2.37539	-2.17486
	ARDL Bull	0.130314	-4.41645	-3.96879
	ARDL Bear	0.260785	-3.75283	-3.32463
Panel D: Best Statistical Model Frequency Combination	ARDL Bull Daily	0.21562	-4.04453	-3.95726
	ARDL Bear Weekly	0.238608	-4.10603	-3.65747
	ARDL Bull Monthly	0.130314	-4.41645	-3.96879

⁴Table II shows the training set MAPE, AIC, and SIC results of each statistical model developed to predict Bitcoin's low price at the daily, weekly, and monthly frequency. The purpose of Table II is to determine the best statistical model frequency combination for predicting Bitcoin's low price. This is done by determining the best statistical model developed to predict Bitcoin's low price at the daily frequency (Panel A), weekly frequency (Panel B), and monthly frequency (Panel C) and then comparing each of those individual frequencies best statistical model training set MAPE, AIC, and SIC to determine the best statistical model frequency combination for predicting Bitcoin's low price (Panel D). Such comparison in Panel D shows that the best statistical model frequency for predicting Bitcoin's low price is the ARDL Bull Monthly model. This is because it yielded the lowest of the three performance metrics (MAPE, AIC, and SIC).

C. Robustness Tests Results of the ARDL Bull Weekly Model Predicting BTC High

The robustness tests results that are presented in Table III below demonstrate that the ARDL Bull Weekly model doesn't solely perform well on its training set but can also perform well on out of sample testing sets that reflect multiple market conditions spanning from 2011 to 2024. This is evident in the robustness tests MAPE range of significantly under the 10% threshold deemed as being a highly accurate forecast by Lewis (1982) with a range of .34% to 3.18% and an average of 1.3% across robustness tests. These findings are indicative of the ARDL Bull Weekly model's ability to adapt to various market conditions when predicting Bitcoin's high price. For a visual representation of these findings please visit Appendix Figures I to VII.

Table III: Out of Sample Tests Results of ARDL Bull Weekly Predicting BTC High

<i>Panel</i>	<i>Test</i>	<i>Sample</i>	<i>MAPE</i>
<i>Panel A: Out-of-Sample Back Tests</i>	2012 Bull Run	August 2012 – December 2013	2.728454
	2013 Bear Run	December 2013 – May 2016	0.961772
	2016 Bull Run	May 2016 – December 2017	0.928219
	2017 Bear Run	December 2017 – March 2020	0.571265
	Full Historical Sample	April 2011 – March 2020	3.184934
<i>Panel B: Out-of-Sample Forward Tests</i>	2021 Bear Run	November 2021 – December 2022	0.342978
	2023 Mini Bull Run	January 2023 – March 2024	0.396297

⁵Table III shows the out of sample back test and forward test MAPE results of the best (in terms of training set MAPE, AIC, and SIC) statistical model frequency combination selected for predicting Bitcoin's high price (ARDL Bull Weekly model). The ARDL Bull Weekly model achieves an astounding MAPE ranging from as low as 0.34% to significantly under the benchmark set by Lewis (1982) in the Journal of Forecasting of 10% at 3.18% in over 10 years of out of sample testing.

D. Robustness Tests Results of the ARDL Bull Monthly Model Predicting BTC Low

Similar to the robustness tests results of the ARDL Bull Weekly model, the robustness tests results of the ARDL Bull Monthly model that are presented in Table IV below indicate that this model also doesn't solely perform well on its training set but can also perform well on out of sample testing sets that reflect multiple market conditions spanning from 2011 to 2024. This is evident in the robustness tests MAPE range of .47% to 6.7% and an average of 3.28% across robustness tests. These findings are indicative of the ARDL Bull Monthly model's ability to adapt to various market conditions when predicting Bitcoin's low price. For a visual representation of these findings please visit Appendix Figures VIII to XIV.

Table IV: Out of Sample Tests Results of ARDL Bull Monthly Predicting BTC Low

<i>Panel</i>	<i>Test</i>	<i>Sample</i>	<i>MAPE</i>
<i>Panel A: Out-of-Sample Back Tests</i>	2012 Bull Run	August 2012 – December 2013	6.052339
	2013 Bear Run	December 2013 – May 2016	4.591585
	2016 Bull Run	May 2016 – December 2017	1.634181
	2017 Bear Run	December 2017 – March 2020	2.024038
	Full Historical Sample	April 2011 – March 2020	6.73458
<i>Panel B: Out-of-Sample Forward Tests</i>	2021 Bear Run	November 2021 – December 2022	1.469782
	2023 Mini Bull Run	January 2023 – March 2024	0.472748

⁶Table IV shows the out of sample back test and forward test MAPE results of the best (in terms of training set MAPE, AIC, and SIC) statistical model frequency combination selected for predicting Bitcoin's low price (ARDL Bull Monthly model). The ARDL Bull Monthly model achieves an astounding MAPE ranging from as low as 0.47% to significantly under the benchmark set by Lewis (1982) in the Journal of Forecasting of 10% at 6.05% in over 10 years of out of sample testing.

E. ARDL Bull Weekly Model and ARDL Bull Monthly Model Performance Comparison to the Performance of the Result Benchmark Papers

As previously displayed in section I. Literature Review, this paper has examined approximately eleven papers within the field of Bitcoin price forecasting. Of those eleven papers, approximately four evaluated their model performance using the same performance metric used to evaluate the ARDL Bull Weekly model and ARDL Bull Monthly model – MAPE. For ease of comparability, these four papers were selected as result benchmark papers.

The purpose of selecting result benchmark papers is to compare their models testing set results to the ARDL Bull Weekly model and ARDL Bull Monthly model testing set results to determine if the latter has a competitive advantage over the literature.

The ARDL Bull Weekly model and ARDL Bull Monthly model as well as the result benchmark papers models yield exceptionally high forecasting results when predicting Bitcoin's price. This is firstly evident in the ARDL Bull Weekly model's average MAPE across robustness tests of 1.3% for predicting Bitcoin's high price and ARDL Bull Monthly model's average MAPE across robustness tests of 3.28% for predicting Bitcoin's low price. This is secondly evident in the result benchmark papers models MAPE of under the 10% threshold as presented in Table V on page 23.

Although the result benchmark papers models have high predictive power of Bitcoin's price, each of the result benchmark papers models were tested on no more than four market conditions, with the majority being tested on only one or two market conditions as presented in Table V. This indicates that these models did not undergo significant comprehensive robustness testing. Such limited robustness testing indicates that these models are both limited in their applicability to other market conditions that are different from the ones that they were tested on and thus their results are more likely the result of chance.

On the contrary, the ARDL Bull Weekly model and ARDL Bull Monthly model are not only accurate and among the models with the lowest MAPE but are also the most comprehensively tested models relative to the literature as presented in Table V. This is evident in their testing set encompassing six market conditions plus Bitcoin's full historical life. This indicates that the ARDL Bull Weekly model and ARDL Bull Monthly model are adaptable and generalizable in their applicability to various market conditions and results are therefore less likely the result of chance. Such high predictive power and adaptability and generalizability across various market conditions relative to the literature indicates that the ARDL Bull Weekly model and ARDL Bull Monthly model have a competitive advantage over the literature and thus, as a contribution to the literature, are among the first to offer traders highly robust forecast for improved risk management.

Table V: ARDL Bull Weekly Model and ARDL Bull Monthly Model Performance Comparison to the Performance of the Result Benchmark Papers

<i>Author(s)</i>	<i>Model</i>	<i>Performance Metric</i>	<i>Result</i>	<i>Testing Set</i>
<i>Im, Ji, and Kim (2020)</i>	LSTM and DNN	MAPE	4.46% and 4.81%	Covered 2 Market Conditions – 2016 Bull Run and 2017 Bear Run
<i>Alazab, Jay, Kalariya, Kumar, Parmar, & Tanwar (2020)</i>	LSTM	MAPE	3.91%	Covered 1 Market Condition – 2017 Bear Run
<i>Abu & Rosbi (2017)</i>	ARIMA	MAPE	5.36%	Covered 1 Market Condition – 2016 Bull Run
<i>Alon, Hassan Shakil & Haque Munim (2019)</i>	ARIMA	MAPE	0.37%	Covered 4 Market Conditions – 2012 Bull Run, 2013 Bear Run, 2016 Bull Run, 2017 Bear Run
<i>Tatiana Rice (2025)</i>	ARDL Bull Weekly and ARDL Bull Monthly	MAPE	1.3% and 3.28%	Covered 6 Market Conditions PLUS Bitcoin’s Historical Life – 2012 Bull Run, 2013 Bear Run, 2016 Bull Run, 2017 Bear Run, 2021 Bear Run, 2023 Mini Bull

⁷Table V shows how this papers MAPE results and robustness testing set compares to prominent papers within the literature that use the same performance metric of MAPE. From the table it is evident that this paper predominately outperforms the literature in terms of MAPE and outperforms the literature in terms of robustness testing. This indicates that this paper is generally more accurate as well as more adaptable and generalizable across various market conditions relative to the literature.

F. Trading Demonstration

F.1. Trading Demonstration for ARDL Bull Weekly Model

Although the ARDL Bull Weekly model has a competitive advantage over the literature, as evident in subsection E of Results and Discussion, can the ARDL Bull Weekly model be used to potentially generate profits? To answer this question, this paper performs the following trading demonstration.

First, to demonstrate the ARDL Bull Weekly model’s profit potential for each of the market conditions included within the out of sample tests, the best performing week for each market condition was selected. Second, for each best performing week, this paper assumes a buy position was entered at Bitcoin’s open price. Third, for each best performing week, this paper also assumes that, using the ARDL Bull Weekly model’s prediction of Bitcoin’s high price, a sell limit was entered.

For simplicity, trading fees were not factored into the profit calculation. Instead, profits were calculated by deducting the entry price – Bitcoin’s open price – from the sell limit price, which is the ARDL Bull Weekly model’s predicted value of Bitcoin’s high price.

Table VI presents the profits generated from this trading demonstration. It shows that if a trader were to trade during the selected weeks for each of the market conditions included within the out of sample tests, using Bitcoin's open price as the entry price and the ARDL Bull Weekly model's predicted value of Bitcoin's high price as the exit price, then if the trader's position size was 1 BTC (Bitcoin), they could have made a profit of as low as \$7.36 and as high as \$610.90. If the trader's position size was 100 BTC, they could have made a profit of as low as \$736 and as high as \$61,090. These findings indicate that the ARDL Bull Weekly model can be used to potentially generate profits.

Table VI: Trading Demonstration Results for ARDL Bull Weekly Model

<i>Sample</i>	<i>Predicted BTC High (Dollar)</i>	<i>Actual BTC High (Dollar)</i>	<i>Actual BTC Open (Dollar)</i>	<i>Profit From Buying 1 BTC</i>	<i>Profit From Buying 100 BTC</i>
<i>2012 Bull Run (Week 50)</i>	97.16	97.50	89.8	7.36	736.00
<i>2013 Bear Run (Week 86)</i>	296.64	296.70	288.7	7.94	794.00
<i>2016 Bull Run (Week 29)</i>	789.04	791.60	774.00	15.04	1,504.00
<i>2017 Bear Run (Week 94)</i>	8,483.96	8,486.20	8,208.00	275.96	27,596.00
<i>Historical Life (Week 443)</i>	8,483.96	8,486.20	8,208.00	275.96	27,596.00
<i>2021 Bear Run (Week 50)</i>	19,679.70	19,692.90	19,068.80	610.90	61,090.00
<i>2023 Mini Bull Run (Week 1)</i>	17,009.32	17,012.30	16,537.50	471.82	47,182.00

⁸Table VI shows the trading demonstration results of the ARDL Bull Weekly model. The results show that for each best performing week across all market conditions Bitcoin has experienced since its inception, had a trader used the ARDL Bull Weekly model to predict Bitcoin's high price they would have made a profit had the trader bought at the actual Bitcoin open price and sold at the ARDL Bull Weekly models prediction of Bitcoin's high price.

F.2. Trading Demonstration for ARDL Bull Monthly Model

Although the ARDL Bull Monthly model also has a competitive advantage over the literature, as evident in subsection E of the Results and Discussion, can the ARDL Bull Monthly model also potentially be used to generate profits? To answer this question, this paper performs another trading demonstration.

First, similar to the trading demonstration performed for the ARDL Bull Weekly model, to demonstrate the profit potential of the ARDL Bull Monthly model for each market condition included within the out of sample tests, the best performing month for each market condition was selected. Second, dissimilar from the trading demonstration performed by the ARDL Bull Weekly model, instead of using Bitcoin's open price as the entry price and predicted values of Bitcoin's high price as the exit price, this paper uses the ARDL Bull Monthly model's predicted

value of Bitcoin low price as the entry price (or buy limit) and uses Bitcoin's close price as the exit price.

For simplicity, trading fees were also not factored into the profit calculation. Instead, profits were calculated by deducting the entry price (or buy limit), which is the ARDL Bull Monthly model's predicted value of Bitcoin's low price, from the exit price – Bitcoin's close price.

Table VII presents the profits generated from this trading demonstration. It shows that if a trader were to trade during the selected months for each of the market conditions included within the out of sample tests, using the ARDL Bull Monthly model's predicted value of Bitcoin's low price as the entry price or buy limit, then if the trader's position size was 1 BTC (Bitcoin), they could have made a profit of as low as \$7.00 and as high as \$2,099.86. If the trader's position size was 100 BTC, they could have made a profit of as low as \$700 and as high as \$209,986. These findings indicate that the ARDL Bull Monthly model can be used to potentially generate profits.

Table VII: Trading Demonstration Results for ARDL Bull Monthly Model

<i>Sample</i>	<i>Predicted BTC Low (Dollar)</i>	<i>Actual BTC Low (Dollar)</i>	<i>Actual BTC Close (Dollar)</i>	<i>Profit From Buying 1 BTC</i>	<i>Profit From Buying 100 BTC</i>
<i>2012 Bull Run (Month 6)</i>	\$13.40	\$13.20	\$20.40	\$7.00	\$700.00
<i>2013 Bear Run (Month 19)</i>	\$221.01	\$221.00	\$264.10	\$43.09	\$4,309.00
<i>2016 Bull Run (Month 8)</i>	\$744.15	\$741.10	\$963.40	\$219.25	\$21,925.00
<i>2017 Bear Run (Month 16)</i>	\$3,725.33	\$3,681.80	\$4,102.30	\$376.97	\$37,697.00
<i>Historical Life (Month 51)</i>	\$221.01	\$221.00	\$264.10	\$43.09	\$4,309.00
<i>2021 Bear Run (Month 12)</i>	\$18,396.44	\$18,207.90	\$20,496.30	\$2,099.86	\$209,986.00
<i>2023 Mini Bull Run (Month 7)</i>	\$29,044.20	\$28,890.70	\$29,232.40	\$188.20	\$18,820.00

⁹Table VII shows the trading demonstration results of the ARDL Bull Monthly model. The results show that for each best performing month across all market conditions Bitcoin has experienced since its inception, had a trader used the ARDL Bull Monthly model to predict Bitcoin's low price they would have made a profit had the trader bought at the ARDL Bull Monthly models predicted value of Bitcoin's low price and sold at the actual Bitcoin close price.

G. Trader Implications

G.1. Improved MAPE Relative to Models Operating Using Dollar Units

Traders can utilize the ARDL Bull Weekly model and the ARDL Bull Monthly model for improved MAPE relative to models operating using traditional trading units – dollar units. This is because both the ARDL Bull Weekly model and the ARDL Bull Monthly model operate using log scale units and such operation enables both models to better detect trends within the data that may not be detected using dollar units. Such enhanced trend detection enables both the ARDL Bull Weekly model and the ARDL Bull Monthly model to yield a lower MAPE of Bitcoin’s high price and Bitcoin’s low price relative to models operating using dollar units.

G.2. Improved Risk Management Across Market Conditions Relative to the Literature

Traders can also utilize the ARDL Bull Weekly model and the ARDL Bull Monthly model to make more informed decisions for improved risk management across market conditions relative to the literature. This is because both the ARDL Bull Weekly model and the ARDL Bull Monthly model yield consistently low MAPE across market conditions spanning over a decade of data and demonstrate superior adaptability and generalizability relative to the literature.

IV. Conclusion

A. Section Overview

To conclude this paper, this section will summarize its main points and then discuss both the limitations and future research that could be executed to reduce these limitations.

B. Summary

The research question of this paper is: Which statistical model frequency-combination best predicts – in terms of MAPE, AIC, and SIC – Bitcoin’s high price and low price. The objective of this paper is to both answer the research question and to ensure that the statistical model-frequency combination selected is both easy to implement and interpret. Answering the research question could benefit traders by enabling them to make more informed decisions for improved risk management. Meeting the objectives could also benefit traders by reducing the probability of worsened risk management caused by implementation and interpretation errors.

To answer the research question and to meet the objectives of this paper, this paper developed two types of statistical models: OLS and ARDL. Three OLS models were created for Bitcoin’s high price (one for each frequency – daily, weekly, and monthly). Three OLS models were also created for Bitcoin’s low price (one also for each frequency). Each OLS model regressed the log transformation of all 23 independent variables against the log transformation of their corresponding dependent variable. The ARDL model consisted of the General ARDL model as well as the ARDL Bull Market model and the ARDL Bear Market model. Three of each of these types of ARDL models were created for Bitcoin’s high price (one for each frequency – daily, weekly, and monthly). This resulted in a total of 9 ARDL models for Bitcoin’s high price. Three of each of these types of ARDL models were also created for Bitcoin’s low price (one also for each frequency). This also resulted in a total of 9 ARDL models for Bitcoin’s low price. Each of these ARDL models regressed the log transformation of Bitcoin’s open price and the log transformation of Bitcoin’s high price or low price against the log transformation of their corresponding dependent variable. In total approximately 24 statistical models were created across multiple frequencies to allow for a comprehensive analysis to answer the research question.

After developing each of these statistical models on a training set and comparing their training set MAPE, AIC, and SIC, the ARDL Bull Weekly model best answered the research question for Bitcoin’s high price and the ARDL Bull Monthly model best answered the research question for Bitcoin’s low price.

Both the ARDL Bull Weekly model and the ARDL Bull Monthly model underwent 7 robustness tests encompassing several out of sample bull markets, bear markets, and Bitcoin’s full historical life, to examine each of these models’ adaptability and generalizability to various market conditions. After examining both models’ robustness tests, both models demonstrated a high level of robustness due to an average MAPE of 1.3% and 3.28% across robustness tests. This is indicative of two models that can adapt to various market conditions to yield accurate predictions of Bitcoin’s high price and Bitcoin’s low price.

Due to these findings, these models could be used by traders to accurately predict Bitcoin’s high price and low price, across various market conditions, so that they can make more informed decisions for improved risk management.

C. Limitations and Future Research

C.1. Overfitting

One minor limitation of the ARDL Bull Weekly model and the ARDL Bull Monthly model is that both models' training set results outperform their testing set results. This indicates overfitting. However, despite the overfitting, as previously discussed, both models perform exceptionally well within the testing set spanning over 10 years in out of sample data. Therefore, although the presence of overfitting is a limitation, it is not a major cause for concern.

C.2. Variable Lags and Retraining

One major limitation is that, because data characteristics can change due to a plethora of reasons that include but aren't limited to changes in market sentiment, regulation, and technology, both the ARDL Bull Weekly and ARDL Bull Monthly models lag structures are likely to change. This means that the lag structure of Bitcoin's high price, low price, and open price within both models should be continuously updated. This way, the models maintain their consistently low MAPE.

Another major limitation is that, because data characteristics can change, both the ARDL Bull Weekly model and the ARDL Bull Monthly model may need to be retrained on an even more recent cryptocurrency bull run as time progresses. This way, the models learn the most recent trends within the data and are able to maintain their consistently low MAPE.

C.3. Over and Underpredictions

Another major limitation of the ARDL Bull Weekly model and the ARDL Bull Monthly model is that because it is not 100% accurate, it occasionally overpredicts and underpredicts. Such over and underpredictions could result in financial loss or missed opportunity. Future research could examine the interaction between the ARDL Bull Weekly model, ARDL Bull Monthly model, and technical indicators to determine if such interaction reduces this overprediction and underprediction problem.

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Appendix

Appendix I: Summary of Bitcoin Price Prediction Studies Using Machine Learning Models

<i>Author(s)</i>	<i>Research Question (Paraphrased)</i>	<i>Methods</i>	<i>Findings</i>
<i>Im, Ji & Kim (2019)</i>	Which machine learning model best predicts Bitcoin's daily spot price between Deep Neural Networks (DNN), Recurrent Neural Networks (RNN), Convolutional Neural Networks (CNN), Resnet, Long Short-Term Memory (LSTM), and Convolutional Recurrent Neural Networks (CRNN)?	Each of these models were trained on a training set spanning from November 2011 to July 2017 using a series of independent variables that were concluded to be Bitcoin price determinants by Bitcoin price determinant studies. These include independent variables such as Bitcoin's block size, mining difficulty, hash rate, market capitalization, miners' revenue, number of transactions, transaction fees, and others.	Relative to the other models in question, the LSTM and DNN are the best predictors of Bitcoin's daily spot price, yielding an MAPE of 4.46% and 4.81%.
<i>Alazab, Jay, Kalariya, Kumar, Parmar & Tanwar (2020)</i>	Which machine learning model best predicts Bitcoin's daily spot price between stochastic Multilayer Perception (MLP), stochastic LSTM, deterministic MLP, and deterministic LSTM?	Each of these models were trained on a training set spanning from 2017 to 2019 and used similar independent variables as used in Im, Ji and Kim (2019).	The stochastic LSTM outperforms the other models within the research question when predicting Bitcoin's daily spot price, yielding an MAPE of 3.91%.
<i>Jiang (2020)</i>	Which machine learning model best predicts Bitcoin's hourly spot price between MLP, LSTM, and Gated Recurrent Unit (GRU)?	Each of these models were trained on a training set spanning from January 2012 to November 2017 and used Bitcoin's price lagged 24 hours as the independent variable.	The GRU best predicts Bitcoin's hourly spot price, yielding a RMSE of 19.02 in dollar units.

¹⁰Appendix I provides a thorough overview (in terms of the research question, methods, and findings) of the Bitcoin price prediction studies discussed within section I. Literature Review, subsection C. This subsection focuses on Bitcoin price prediction studies that predict Bitcoin's price using solely machine learning models. From these studies it is evident that machine learning models, particularly LSTM and GRU, show promise at forecasting Bitcoin's price due to yielding low error rates of an MAPE under the 10% benchmark set by Lewis (1982) in the Journal of Forecasting and RMSE that represents less than 1% of the value of Bitcoin's price.

Appendix II: Summary of Bitcoin Price Prediction Studies Using Statistical Models

<i>Author(s)</i>	<i>Research Question (Paraphrased)</i>	<i>Methods</i>	<i>Findings</i>
<i>Chilamkurti, M & V (2020)</i>	Can the Autoregressive Integrated Moving Average (ARIMA) predict Bitcoin's daily closing price and Bitcoin's hourly closing price?	To answer the first half of the research question, the ARIMA model was trained on a training set spanning from April 2013 to December 2016 and used the lags of Bitcoin's daily closing price as the independent variable. To answer the second half of the research question, the ARIMA model was trained on the same training set and used lags of Bitcoin's hourly closing price as the independent variable.	The ARIMA model can predict Bitcoin's daily closing price, achieving an accuracy rate of 49%. However, the ARIMA model cannot predict Bitcoin's hourly closing price.
<i>Abu & Rosbi (2017)</i>	Can ARIMA predict Bitcoin's monthly spot price?	The ARIMA model was trained on a training set spanning from January 2013 to August 2017 using 2 lags of Bitcoin's monthly spot price and 2 lags of the moving average as the independent variables.	The ARIMA model can predict Bitcoin's monthly spot price with high accuracy with a MAPE of 5.36%.
<i>Kuen Siu (2022)</i>	Which statistical model between Autoregressive Self Exciting Threshold (AR-Setar), Generalized Autoregressive Conditional Heteroskedasticity (GARCH), and Stochastic Volatility (SV), best predict Bitcoin's daily closing price under different states of volatility?	Each of the three models were trained on a training set spanning from January 2018 to December 2020 using the lag of Bitcoin's daily closing price as the independent variable.	The SV model best predicts Bitcoin's daily closing price under different states of volatility relative to the AR-Setar and GARCH models, achieving a log likelihood of -3443.83.

¹¹ Appendix II provides a thorough overview (in terms of the research question, methods, and findings) of the Bitcoin price prediction studies discussed within section I. Literature Review, subsection D. This subsection focuses on Bitcoin price prediction studies that predict Bitcoin's price using solely statistical models. From these studies it is evident that machine learning models, particularly ARIMA and SV, show promise at forecasting Bitcoin's price due to yielding low error rates of an MAPE under the 10% benchmark set by Lewis (1982) in the Journal of Forecasting and negative Log Likelihood that indicates that the models predicted values of Bitcoin's price closely match the actual values of Bitcoin's price.

Appendix III: Summary of Bitcoin Price Prediction Studies: Machine Learning & Statistical

<i>Author(s)</i>	<i>Research Question (Paraphrased)</i>	<i>Methods</i>	<i>Findings</i>
<i>Chen, Li & Sun (2019)</i>	Which model between statistical models Logistic Regression and Linear Discriminant Analysis and machine learning models Random Forest, XGBoost, Quadratic Discriminant Analysis, Support Vector Machine, and LSTM best predicts Bitcoin's daily spot price and best predicts Bitcoin's 5-minute spot price?	To answer the first half of this research question, each of these models were trained on a training set spanning from February 2017 to February 2019 using similar independent variables that were used in Im, Ji, and Kim (2019). To answer the second half of this research question, each of these models were trained on a training set spanning from July 2017 to January 2018, also using similar independent variables as used in Im, Ji and Kim (2019).	The statistical model Logistic Regression best predicts Bitcoin's daily spot price with an accuracy rate of 66%. On the contrary, the machine learning model LSTM best predicts Bitcoin's 5-minute spot price with an accuracy rate of 67.2%.
<i>Ho, Kumar & Vatambeti (2021)</i>	Which model between statistical model Linear Regression and machine learning model LSTM best predicts Bitcoin's daily spot price?	Each of these models were trained on a training set spanning from August 2017 to August 2020 using Bitcoin's daily close price, open price, high price, and low price as independent variables.	The machine learning model LSTM best predicts Bitcoin's daily spot price relative to the statistical model Linear Regression.
<i>Daniel, Olorunfemi & Oluwatobi (2022)</i>	Which model between statistical models Linear Regression and ARIMA and machine learning models Naïve, Exponential Smoothing, Neural Network, STL, and Holt Winters Filer best predicts Bitcoin's daily, weekly, and monthly spot price?	Each of these models were trained on a training set spanning from January 2015 to November 2021 using the lags of the dependent variable as the independent variables.	The machine learning model Naïve best predicts Bitcoin's daily spot price while the statistical model Linear Regression best predicts Bitcoin's weekly and monthly spot price.
<i>Alon, Hassan Shakil & Haque Munim (2019)</i>	Which model between statistical model ARIMA and machine learning model Neural Network Autoregressive (NNAR) best predicts Bitcoin's daily spot price?	Each of these models were trained on a training set spanning from January 2012 to June 2017 using lags of Bitcoin's daily spot price as the independent variable.	The ARIMA model best predicts Bitcoin's daily spot price relative to NNAR, yielding an MAPE of .37%.
<i>Tripathi and Tripathi (2022)</i>	Which model between statistical model ARIMA and machine learning models Random Walk (RW), Artificial Neural Network (ANN), and the Ensemble Model best predicts Bitcoin's daily spot price?	Each of these models were trained on a training set spanning from April 2013 to June 2020 using lags of Bitcoin's daily spot price as the independent variable.	The ensemble model best predicts Bitcoin's daily spot price relative to the statistical model ARIMA and the machine learning models RW and ANN.

¹²Appendix III provides a thorough overview (in terms of the research question, methods, and findings) of the Bitcoin price prediction studies discussed within section I. Literature Review, subsection E. This subsection focuses on Bitcoin price prediction studies that predict Bitcoin's price using both machine learning and statistical models. From these studies it is evident that while machine learning models and statistical models show promise at forecasting Bitcoin's price, machine learning models are more accurate than statistical models when forecasting Bitcoin's price at high frequency and statistical models are more accurate than machine learning models when forecasting Bitcoin's price at low frequency.

Appendix IV: Independent Variable Definitions

<i>Endogenous Variables</i>	<i>Definition</i>	<i>Exogenous Variables</i>	<i>Definition</i>
<i>Bitcoin Open Price</i>	The price of Bitcoin at the beginning of a specified time period.	<i>U.S Federal Funds Rate</i>	The interest rate at which depository institutions lend reserve balances to other depository institutions overnight on an uncollateralized basis.
<i>Bitcoin Stripped Blocksize</i>	The size of a Bitcoin block after removing certain non-essential data.	<i>U.S GDP</i>	The total monetary or market value of all final goods and services produced in a country in a specific time period.
<i>Bitcoin Mining Difficulty</i>	A measure of how difficult it is to find a hash below a given target.	<i>U.S CPI Inflation Rate</i>	The percentage change in the Consumer Price Index (CPI) from one period to the next.
<i>Bitcoin Hash Rate</i>	The computational power of the Bitcoin network.	<i>U.S M2 Money Supply</i>	A measure of the money supply that includes cash, checking deposits, and easily convertible near money.
<i>Bitcoin Market Capitalization</i>	The total dollar value of all Bitcoins in circulation.	<i>Dow Jones Index Open Price</i>	The price level of the Dow Jones Industrial Average at the beginning of a specified time period.
<i>Bitcoin Miners Revenue</i>	The total income earned by Bitcoin miners.	<i>Gold Open Price</i>	The price of gold at the beginning of a specified time period.
<i>Bitcoin Number of Transactions</i>	The total count of transactions recorded on the Bitcoin blockchain.	<i>Gold Open Volatility</i>	The degree of variation in the price of gold over time.
<i>Bitcoin Circulating Supply</i>	The total number of Bitcoins in circulation.	<i>Oil Open Price</i>	The price of oil at the beginning of a specified time period.
<i>Bitcoin Total Transaction Fees</i>	The cumulative amount of fees paid by users for transactions on the Bitcoin network.	<i>Oil Open Volatility</i>	The degree of variation in the price of oil over time.
<i>Bitcoin Price Volatility</i>	The degree of variation in the price of Bitcoin over time.	<i>S&P500 Index Open Price</i>	The price level of the S&P 500 index at the beginning of a specified time period.
		<i>U.S 10 Year Bond Open Yield</i>	The interest rate for U.S. Treasury bonds with a maturity of 10 years at the beginning of a specified time period.
		<i>USD Index Open Price</i>	The value of the U.S. dollar relative to a basket of foreign currencies at the beginning of a specified time period.
		<i>Google Trend</i>	A measure of the popularity of a search term relative to its total search volume on Google.

¹³Appendix IV provides the definitions for the endogenous and exogenous independent variables used within this paper. The endogenous variables and exogenous variables were generated by ChatGPT based on commonly accepted definitions in the field. Each of the endogenous variable's definitions are based on descriptions provided by cryptocurrency market analysis platforms such as CoinMarketCap, Binance, and Blockchain.com. The U.S Federal Funds Rate, U.S GDP, U.S CPI Inflation Rate, and U.S M2 Money Supply definitions are based on economic reports from the Federal Reserve and the Bureau of Economic Analysis. The Dow Jones Index Open Price, Gold Open Price, Gold Open Volatility, Oil Open Price, Oil Open Volatility, and S&P 500 Index Open Price definitions are sourced from financial market data providers such as Bloomberg and Reuters. The U.S 10 Year Bond Open Yield definition is sourced from U.S Treasury reports, and the USD Index Open Price is based on foreign exchange market data from financial analysis platforms. Lastly, Google Trend is based on information provided by Google Trends.

Appendix V: Independent Variable Data Sources

<i>Endogenous Variables</i>			<i>Exogenous Variables</i>		
<i>Investing.com</i>	<i>Bitcoinvisuals.com</i>	<i>Data.Bitcoinity.com</i>	<i>FRED</i>	<i>Investing.com</i>	<i>Google Search Engine</i>
Bitcoin Open Price	Bitcoin Stripped Blocksize	Bitcoin Price Volatility	U.S Federal Funds Rate	Dow Jones Index Open Price	Google Trend
	Bitcoin Mining Difficulty		U.S GDP	Gold Open Price	
	Bitcoin Hash Rate		U.S CPI Inflation Rate	Gold Open Volatility	
	Bitcoin Market Capitalization		U.S M2 Money Supply	Oil Open Price	
	Bitcoin Miners Revenue			Oil Open Volatility	
	Bitcoin Number of Transactions			S&P500 Index Open Price	
	Bitcoin Circulating Supply			U.S 10 Year Bond Open Yield	
	Bitcoin Total Transaction Fees			USD Index Open Price	

¹⁴Appendix V provides the data sources for the endogenous and exogenous independent variables used within this paper.

Appendix VI: Dependent Variable Definitions and Data Sources

<i>Dependent Variables</i>	<i>Definition</i>	<i>Source</i>
Bitcoin High Price	The highest recorded price of Bitcoin during a specified time period.	Investing.com
Bitcoin Low Price	The lowest recorded price of Bitcoin during a specified time period.	Investing.com

¹⁵Appendix VI provides the definitions and data sources for the dependent variables used within this paper.

Appendix VII: Table Representation of OLS Developed for Bitcoin's High and Low Price

<i>Frequency</i>	<i>Dependent Variable</i>	<i>Independent Variables</i>	<i>Transformation Applied</i>
Daily	Bitcoin's High Price	23 variables	Log-Log
Weekly	Bitcoin's High Price	23 variables	Log-Log
Monthly	Bitcoin's High Price	23 variables	Log-Log
Daily	Bitcoin's Low Price	23 variables	Log-Log
Weekly	Bitcoin's Low Price	23 variables	Log-Log
Monthly	Bitcoin's Low Price	23 variables	Log-Log

¹⁶Appendix VII provides a visual table representation of the OLS models developed to predict Bitcoin's high price and low price at the daily, weekly, and monthly frequency.

Appendix VIII: Table Representation of General ARDL for Bitcoin's High and Low Price

<i>Frequency</i>	<i>Dependent Variables</i>	<i>Independent Variables</i>	<i>Transformation</i>	<i>Lag Structure</i>
Daily	Bitcoin's high price	Bitcoin's open price	Log-Log	(1,8)
Weekly	Bitcoin's high price	Bitcoin's open price	Log-Log	(2,15)
Monthly	Bitcoin's high price	Bitcoin's open price	Log-Log	(1,2)
Daily	Bitcoin's low price	Bitcoin's open price	Log-Log	(6,5)
Weekly	Bitcoin's low price	Bitcoin's open price	Log-Log	(2,15)
Monthly	Bitcoin's low price	Bitcoin's open price	Log-Log	(5,2)

¹⁷Appendix VIII provides a visual table representation of the General ARDL models developed to predict Bitcoin's high price and low price at the daily, weekly, and monthly frequency.

Appendix IX: Table Representation of ARDL Bull Models for Bitcoin's High and Low Price

<i>Frequency</i>	<i>Dependent Variables</i>	<i>Independent Variables</i>	<i>Transformation</i>	<i>Lag Structure</i>
Daily	Bitcoin's high price	Bitcoin's open price	Log-Log	(1,8)
Weekly	Bitcoin's high price	Bitcoin's open price	Log-Log	(2,15)
Monthly	Bitcoin's high price	Bitcoin's open price	Log-Log	(1,2)
Daily	Bitcoin's low price	Bitcoin's open price	Log-Log	(6,5)
Weekly	Bitcoin's low price	Bitcoin's open price	Log-Log	(2,15)
Monthly	Bitcoin's low price	Bitcoin's open price	Log-Log	(5,2)

¹⁸Appendix IX provides a visual table representation of the ARDL Bull models developed to predict Bitcoin's high price and low price at the daily, weekly, and monthly frequency.

Appendix X: Table Representation of ARDL Bear Models for Bitcoin's High and Low Price

<i>Frequency</i>	<i>Dependent Variables</i>	<i>Independent Variables</i>	<i>Transformation</i>	<i>Lag Structure</i>
Daily	Bitcoin's high price	Bitcoin's open price	Log-Log	(1,8)
Weekly	Bitcoin's high price	Bitcoin's open price	Log-Log	(2,15)
Monthly	Bitcoin's high price	Bitcoin's open price	Log-Log	(1,2)
Daily	Bitcoin's low price	Bitcoin's open price	Log-Log	(6,5)
Weekly	Bitcoin's low price	Bitcoin's open price	Log-Log	(2,15)
Monthly	Bitcoin's low price	Bitcoin's open price	Log-Log	(5,2)

¹⁹Appendix X provides a visual table representation of the ARDL Bear models developed to predict Bitcoin's high price and low price at the daily, weekly, and monthly frequency.

Appendix XI: Table Representation of Bitcoin High Price Model Comparison Per Frequency

<i>Frequency</i>	<i>Model Type</i>	<i>Performance Metric and Criteria</i>
Daily	OLS	MAPE, AIC, SIC
	General ARDL (1,8)	MAPE, AIC, SIC
	ARDL Bull (1,8)	MAPE, AIC, SIC
	ARDL Bear (1,8)	MAPE, AIC, SIC
Weekly	OLS	MAPE, AIC, SIC
	General ARDL (2,15)	MAPE, AIC, SIC
	ARDL Bull (2,15)	MAPE, AIC, SIC
	ARDL Bear (2,15)	MAPE, AIC, SIC
Monthly	OLS	MAPE, AIC, SIC
	General ARDL (1,2)	MAPE, AIC, SIC
	ARDL Bull (1,2)	MAPE, AIC, SIC
	ARDL Bear (1,2)	MAPE, AIC, SIC

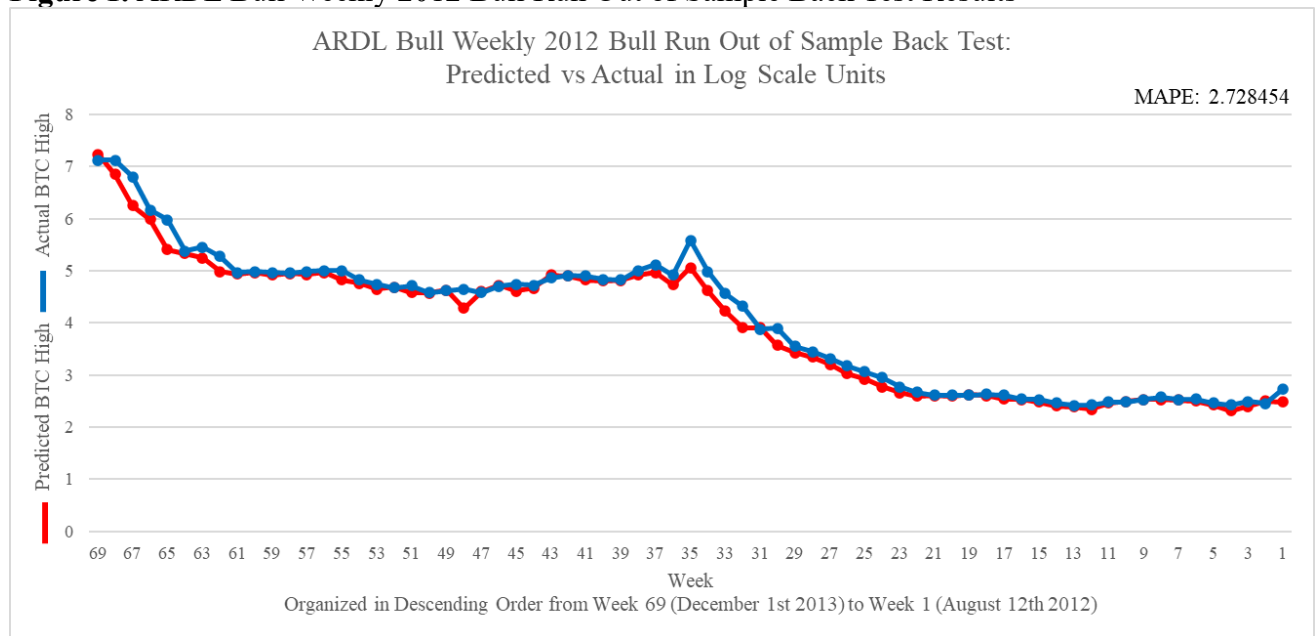
²⁰Appendix XI provides a visual table representation of how for a given frequency, each statistical model that was developed to predict Bitcoin's high price was compared in terms of training set MAPE, AIC, and SIC to ultimately determine the best statistical model for that given frequency. This was done in order to ultimately compare the best statistical model for the daily, weekly, and monthly frequency, to determine which of the three statistical models yield the lowest training set MAPE, AIC, and SIC and thus is the best statistical model frequency combination for predicting Bitcoin's high price.

Appendix XII: Table Representation of Bitcoin Low Price Model Comparison Per Frequency

<i>Frequency</i>	<i>Model Type</i>	<i>Performance Metric and Criteria</i>
Daily	OLS	MAPE, AIC, SIC
	General ARDL (6,5)	MAPE, AIC, SIC
	ARDL Bull (6,5)	MAPE, AIC, SIC
	ARDL Bear (6,5)	MAPE, AIC, SIC
Weekly	OLS	MAPE, AIC, SIC
	General ARDL (2,15)	MAPE, AIC, SIC
	ARDL Bull (2,15)	MAPE, AIC, SIC
	ARDL Bear (2,15)	MAPE, AIC, SIC
Monthly	OLS	MAPE, AIC, SIC
	General ARDL (5,2)	MAPE, AIC, SIC
	ARDL Bull (5,2)	MAPE, AIC, SIC
	ARDL Bear (5,2)	MAPE, AIC, SIC

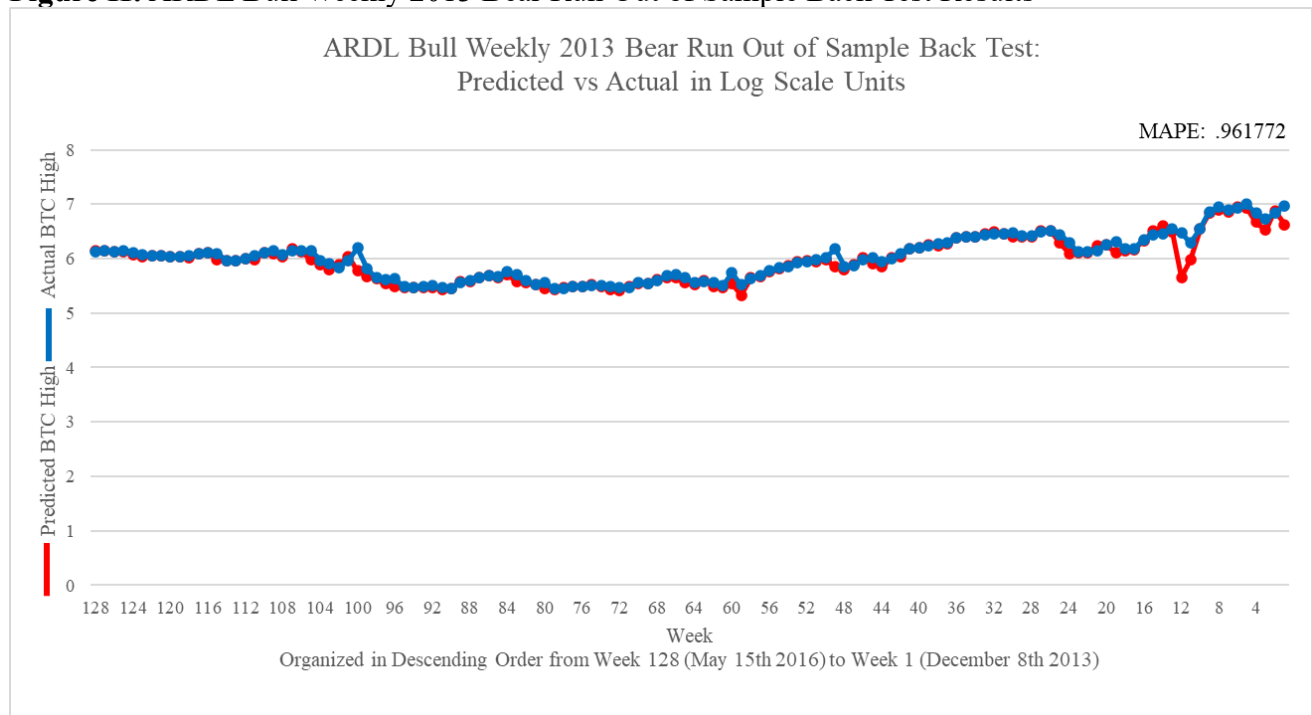
²¹Appendix XII provides a visual table representation of how for a given frequency, each statistical model that was developed to predict Bitcoin's low price was compared in terms of training set MAPE, AIC, and SIC to ultimately determine the best statistical model for that given frequency. This was done in order to ultimately compare the best statistical model for the daily, weekly, and monthly frequency, to determine which of the three statistical models yield the lowest training set MAPE, AIC, and SIC and thus is the best statistical model frequency combination for predicting Bitcoin's low price.

Figure I: ARDL Bull Weekly 2012 Bull Run Out of Sample Back Test Results



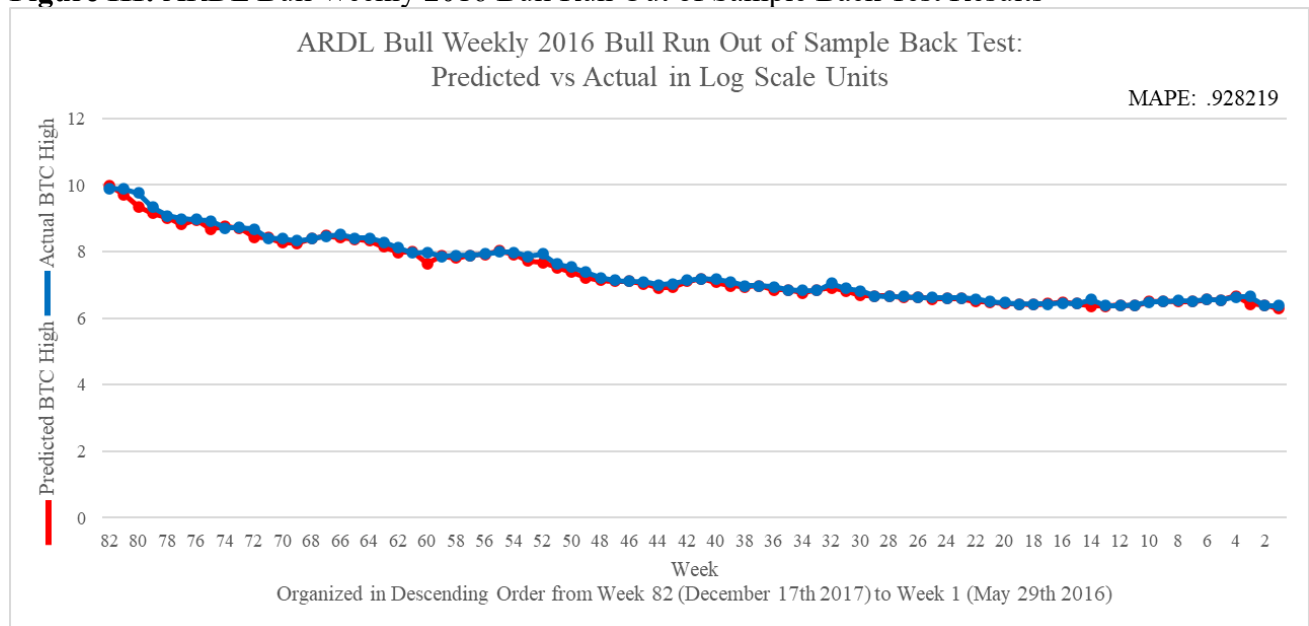
22 Figure I provides a visual graphical representation of the 2012 bull run out of sample back test MAPE results of the ARDL Bull Weekly model for predicting Bitcoin's high price. The graph moves from left to right in descending weeks starting with the very last week of the 2012 bull run (week 69 or December 1, 2013) and ending with the very first week of the 2012 bull run (week 1 or August 12, 2012). The red line represents the ARDL Bull Weekly models predicted value of Bitcoin's high price, and the blue line represents the actual value of Bitcoin's high price. Based on the graph, it is evident that the ARDL Bull Weekly model performed exceptionally well during the 2012 bull run. This is due to its predicted values proximity to the actual value throughout the 2012 bull run as evident by the red and blue lines as well as its low MAPE of 2.7% which is significantly below the 10% benchmark set by Lewis (1982) in the Journal of Forecasting.

Figure II: ARDL Bull Weekly 2013 Bear Run Out of Sample Back Test Results



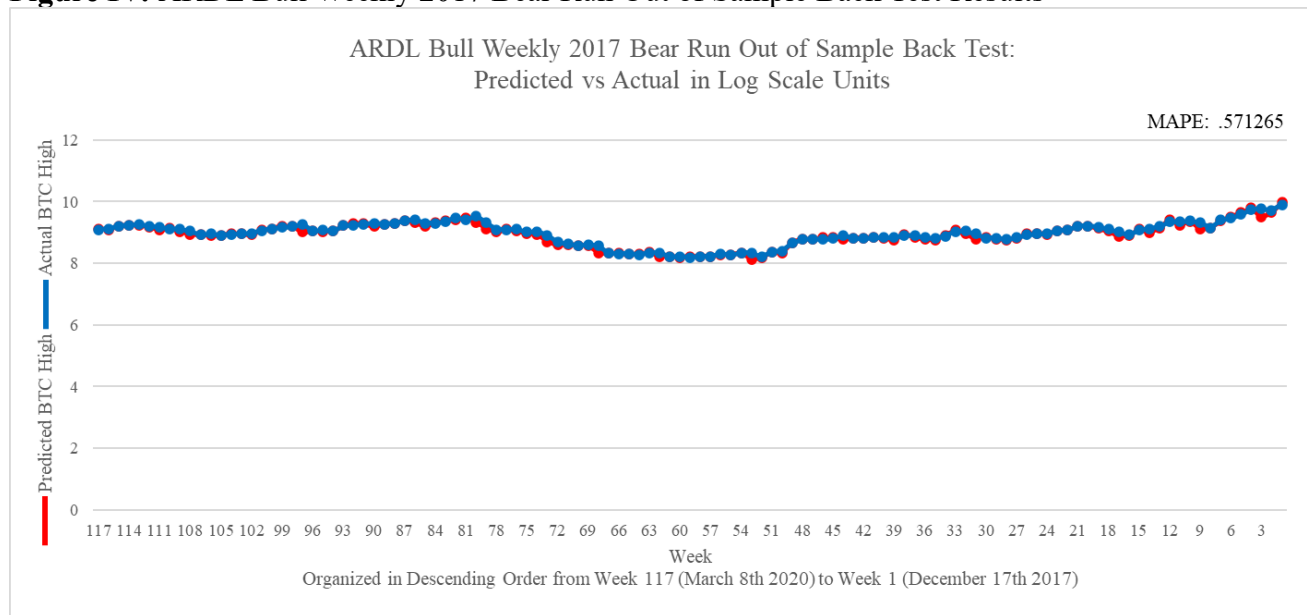
²³ Figure II provides a visual graphical representation of the 2013 bear run out of sample back test MAPE results of the ARDL Bull Weekly model for predicting Bitcoin's high price. The graph moves from left to right in descending weeks starting with the very last week of the 2013 bear run (week 128 or May 15, 2016) and ending in the very first week of the 2013 bear run (week 1 or December 8, 2013). The red line represents the ARDL Bull Weekly models predicted value of Bitcoin's high price, and the blue line represents the actual value of Bitcoin's high price. Based on the graph, it is evident that the ARDL Bull Weekly model performed impressively well during the 2013 bear run. This is due to its predicted values proximity to the actual value throughout the 2013 bear run as evident by the red and blue lines as well as its low MAPE of 0.96% which is significantly below the 10% benchmark set by Lewis (1982) in the Journal of Forecasting.

Figure III: ARDL Bull Weekly 2016 Bull Run Out of Sample Back Test Results



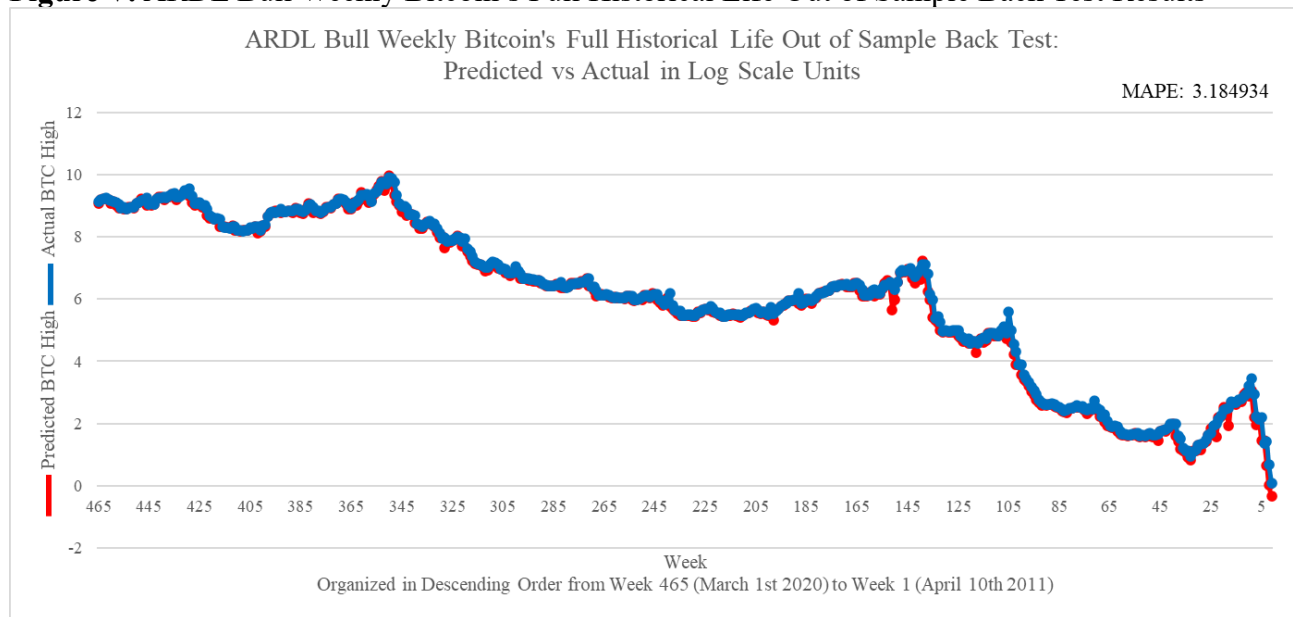
²⁴ Figure III provides a visual graphical representation of the 2016 bull run out of sample back test MAPE results of the ARDL Bull Weekly model for predicting Bitcoin's high price. The graph moves from left to right in descending weeks starting with the very last week of the 2016 bull run (week 82 or December 17, 2017) and ending in the very first week of the 2016 bull run (week 1 or May 29, 2016). The red line represents the ARDL Bull Weekly models predicted value of Bitcoin's high price, and the blue line represents the actual value of Bitcoin's high price. Based on the graph, it is evident that the ARDL Bull Weekly model performed very well during the 2016 bull run. This is due to its predicted values proximity to the actual value throughout the 2016 bull run as evident by the red and blue lines as well as its low MAPE of 0.92% which is significantly below the 10% benchmark set by Lewis (1982) in the Journal of Forecasting.

Figure IV: ARDL Bull Weekly 2017 Bear Run Out of Sample Back Test Results



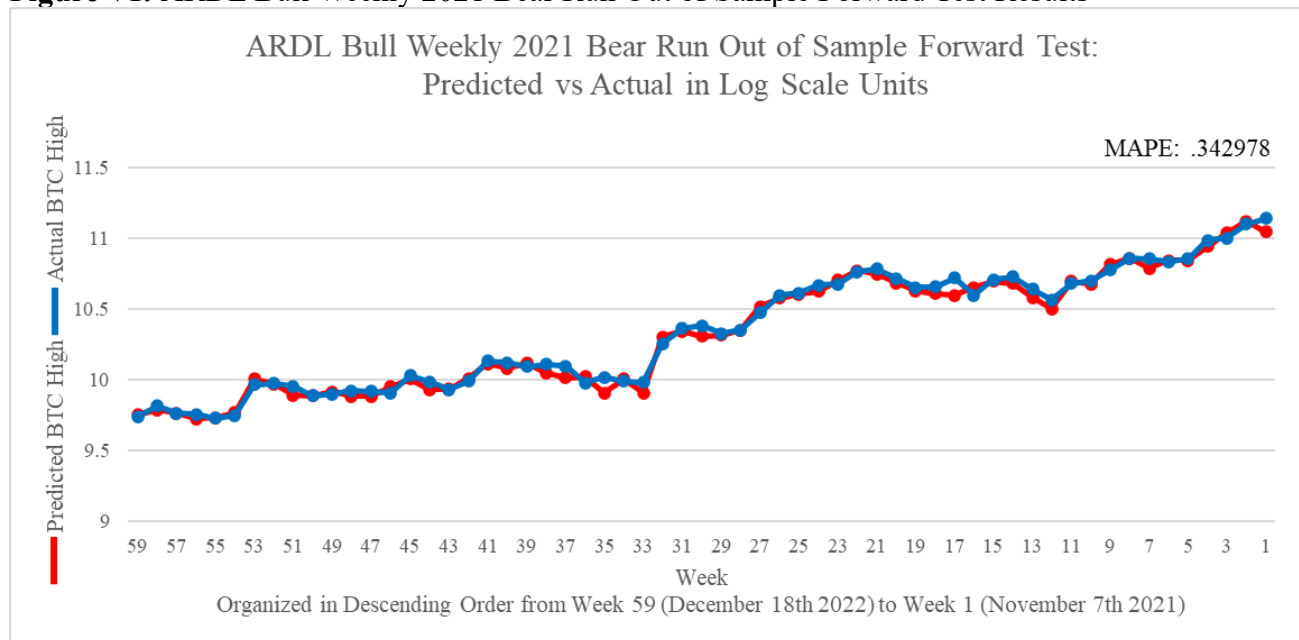
²⁵ Figure IV provides a visual graphical representation of the 2017 bear run out of sample back test MAPE results of the ARDL Bull Weekly model for predicting Bitcoin's high price. The graph moves from left to right in descending weeks starting with the very last week of the 2017 bear run (week 117 or March 8, 2020) and ending in the very first week of the 2017 bear run (week 1 or December 17, 2017). The red line represents the ARDL Bull Weekly models predicted value of Bitcoin's high price, and the blue line represents the actual value of Bitcoin's high price. Based on the graph, it is evident that the ARDL Bull Weekly model performed very well during the 2017 bear run. This is due to its predicted values proximity to the actual value throughout the 2017 bear run as evident by the red and blue lines as well as its low MAPE of 0.57% which is significantly below the 10% benchmark set by Lewis (1982) in the Journal of Forecasting.

Figure V: ARDL Bull Weekly Bitcoin's Full Historical Life Out of Sample Back Test Results



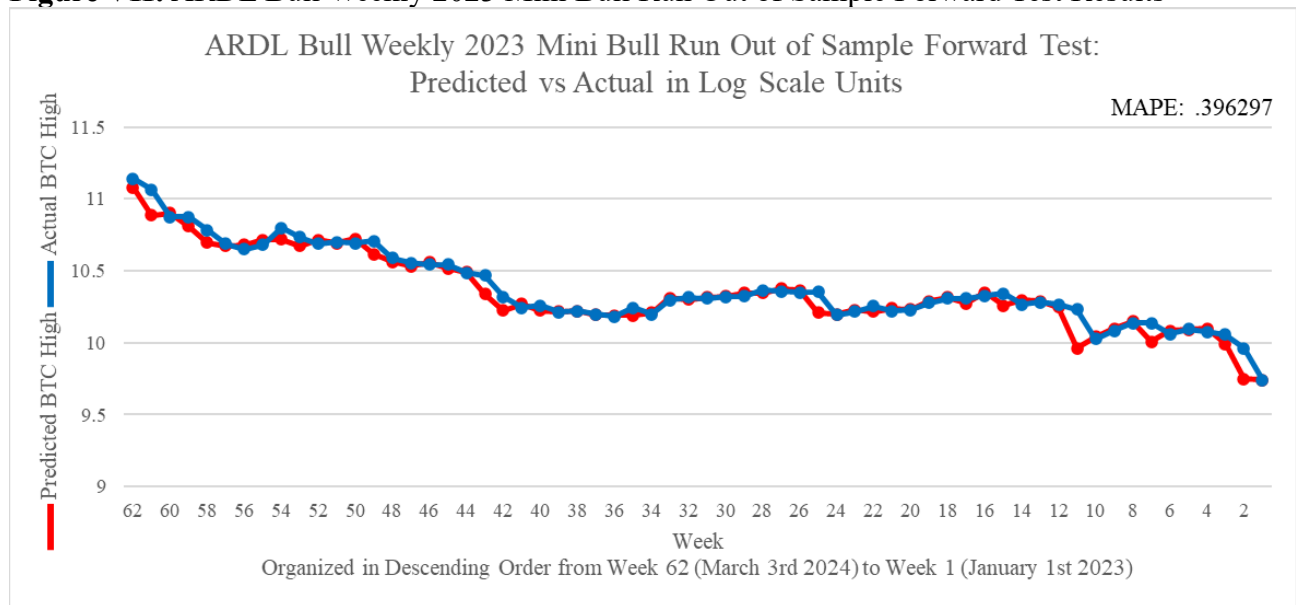
²⁶ Figure V provides a visual graphical representation of Bitcoin's full historical life out of sample back test MAPE results of the ARDL Bull Weekly model for predicting Bitcoin's high price. The graph moves from left to right in descending weeks starting with the last week of Bitcoin's full historical life (week 465 or March 1, 2020) and ending in the first week of Bitcoin's full historical life (week 1 or April 10, 2011). The red line represents the ARDL Bull Weekly models predicted value of Bitcoin's high price, and the blue line represents the actual value of Bitcoin's high price. Based on the graph, it is evident that the ARDL Bull Weekly model performed very well during Bitcoin's full historical life. This is due to its predicted values proximity to the actual value throughout Bitcoin's full historical life as evident by the red and blue lines as well as its low MAPE of 3.18% which is significantly below the 10% benchmark set by Lewis (1982) in the Journal of Forecasting.

Figure VI: ARDL Bull Weekly 2021 Bear Run Out of Sample Forward Test Results



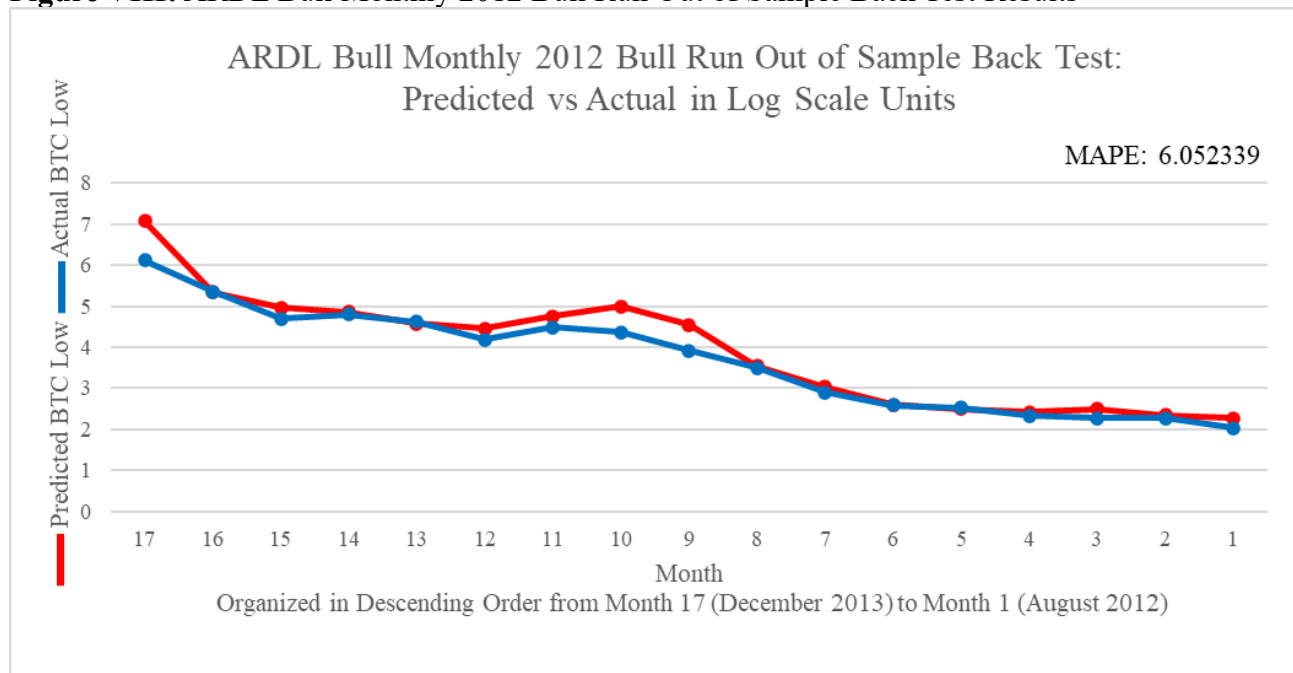
²⁷ Figure VI provides a visual graphical representation of the 2021 bear run out of sample forward test MAPE results of the ARDL Bull Weekly model for predicting Bitcoin's high price. The graph moves from left to right in descending weeks starting with the very last week of the 2021 bear run (week 59 or December 18, 2022) and ending in the very first week of the 2021 bear run (week 1 or November 7, 2021). The red line represents the ARDL Bull Weekly models predicted value of Bitcoin's high price, and the blue line represents the actual value of Bitcoin's high price. Based on the graph, it is evident that the ARDL Bull Weekly model performed very well during the 2021 bear run. This is due to its predicted values proximity to the actual value throughout the 2021 bear run as evident by the red and blue lines as well as its low MAPE of 0.34% which is significantly below the 10% benchmark set by Lewis (1982) in the Journal of Forecasting.

Figure VII: ARDL Bull Weekly 2023 Mini Bull Run Out of Sample Forward Test Results



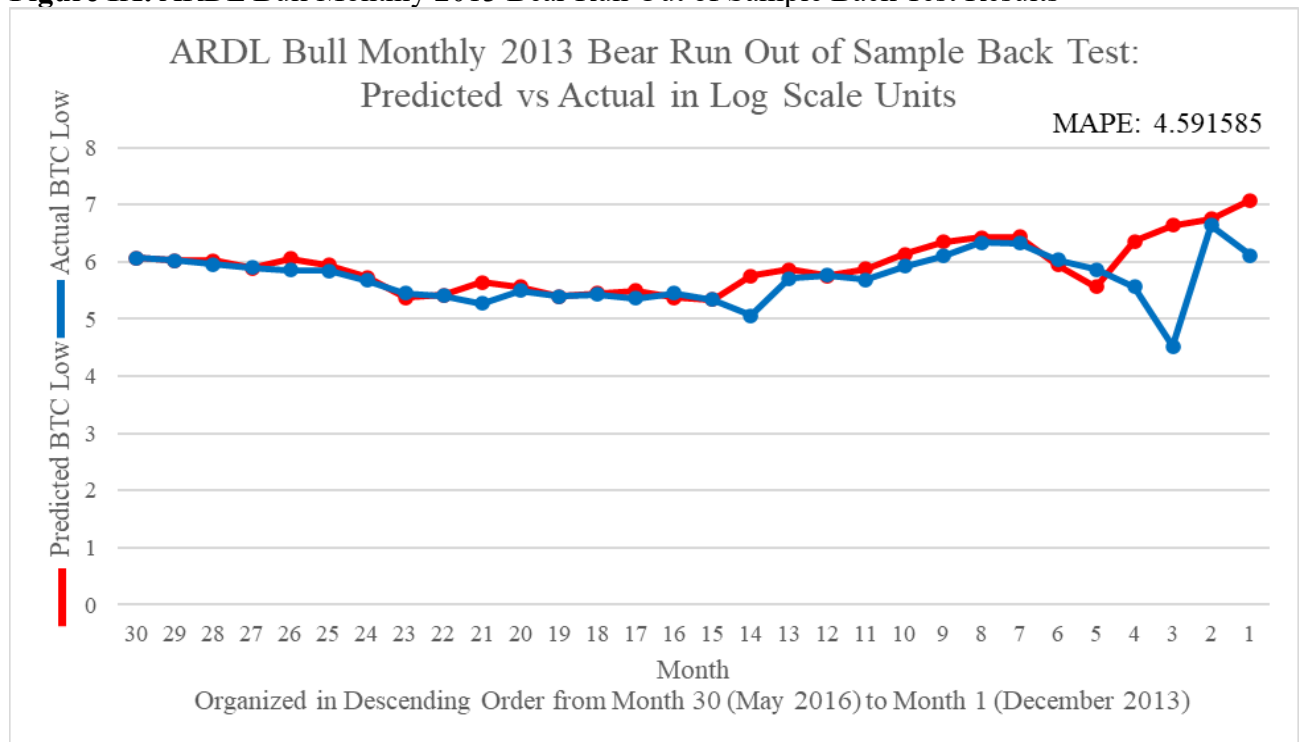
²⁸ Figure VII provides a visual graphical representation of the 2023 mini bull run out of sample forward test MAPE results of the ARDL Bull Weekly model for predicting Bitcoin's high price. The graph moves from left to right in descending weeks starting with the very last week of the 2023 mini bull run (week 62 or March 3, 2024) and ending in the very first week of the 2023 mini bull run (week 1 or January 1, 2023). The red line represents the ARDL Bull Weekly models predicted value of Bitcoin's high price, and the blue line represents the actual value of Bitcoin's high price. Based on the graph, it is evident that the ARDL Bull Weekly model performed very well during the 2023 mini bull run. This is due to its predicted values proximity to the actual value throughout the 2023 mini bull run as evident by the red and blue lines as well as its low MAPE of 0.39% which is significantly below the 10% benchmark set by Lewis (1982) in the Journal of Forecasting.

Figure VIII: ARDL Bull Monthly 2012 Bull Run Out of Sample Back Test Results



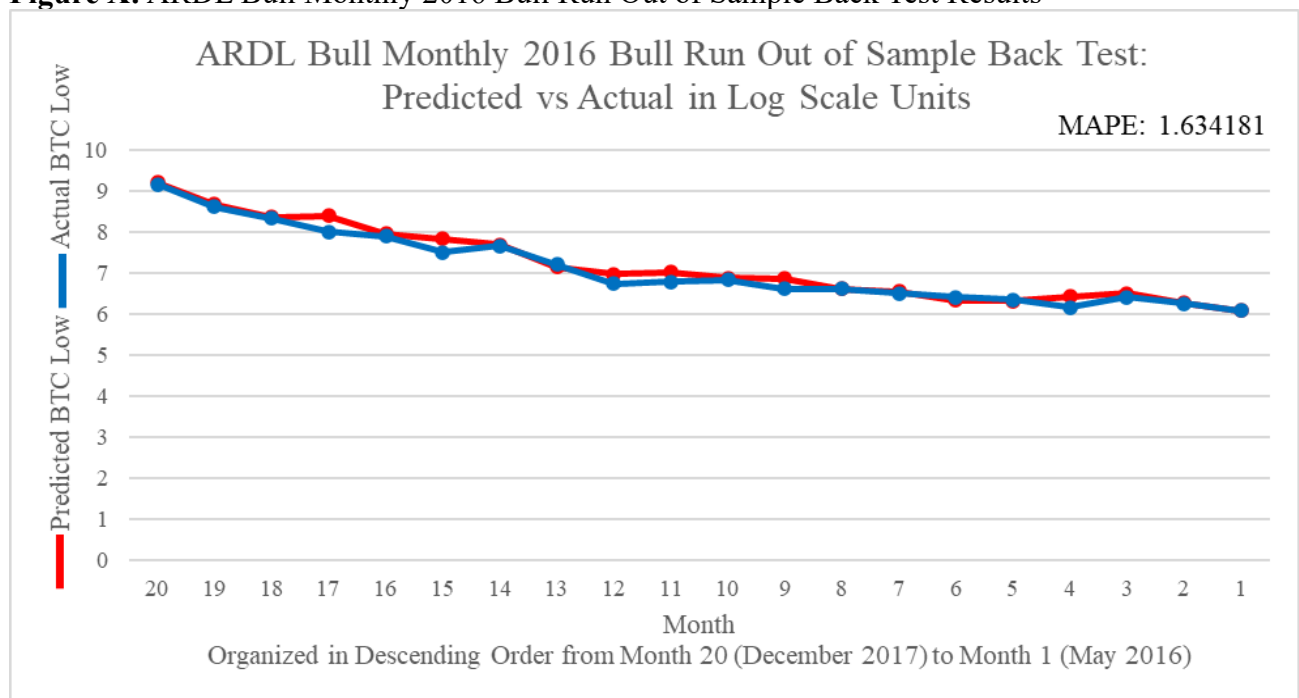
²⁹ Figure VIII provides a visual graphical representation of the 2012 bull run out of sample back test MAPE results of the ARDL Bull Monthly model for predicting Bitcoin's low price. The graph moves from left to right in descending months starting with the very last month of the 2012 bull run (month 17 or December 2013) and ending with the very first month of the 2012 bull run (month 1 or August 2012). The red line represents the ARDL Bull Monthly models predicted value of Bitcoin's low price, and the blue line represents the actual value of Bitcoin's low price. Based on the graph, it is evident that the ARDL Bull Monthly model performed exceptionally well during the 2012 bull run. This is due to its predicted values proximity to the actual value throughout the 2012 bull run as evident by the red and blue lines as well as its low MAPE of 6.0% which is significantly below the 10% benchmark set by Lewis (1982) in the Journal of Forecasting.

Figure IX: ARDL Bull Monthly 2013 Bear Run Out of Sample Back Test Results



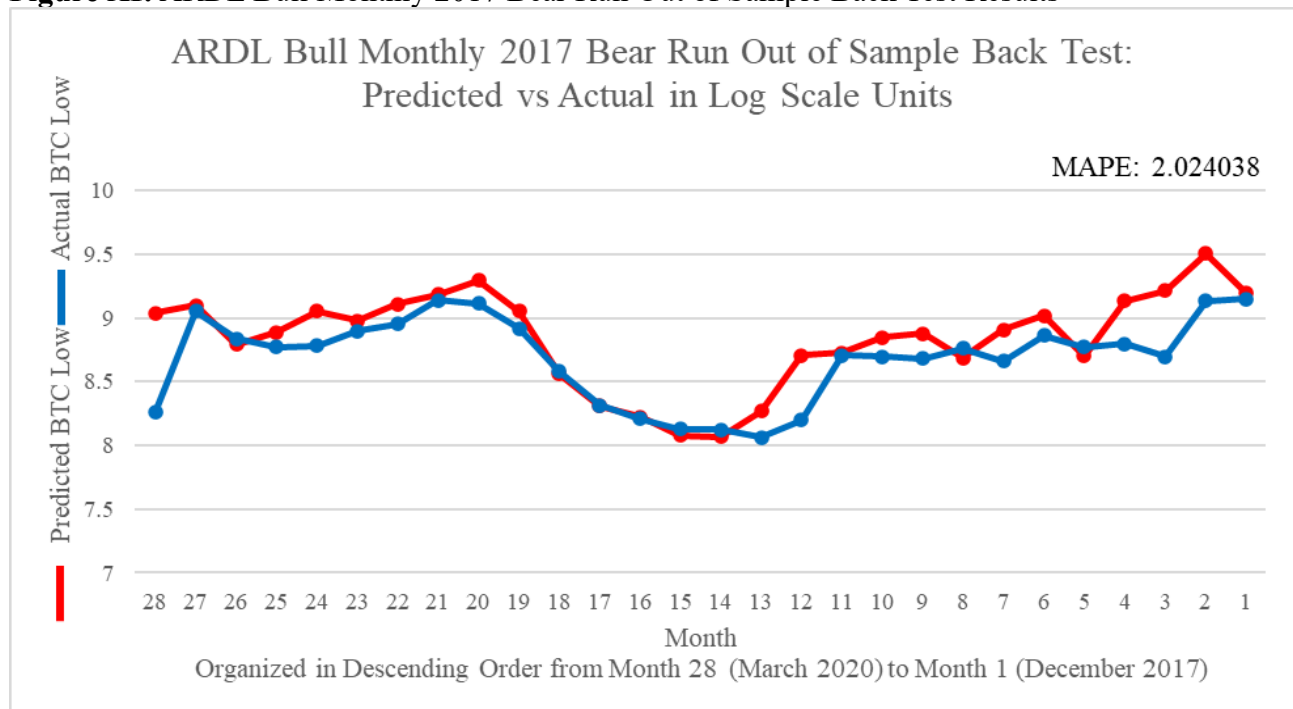
³⁰ Figure IX provides a visual graphical representation of the 2013 bear run out of sample back test MAPE results of the ARDL Bull Monthly model for predicting Bitcoin's low price. The graph moves from left to right in descending months starting with the very last month of the 2013 bear run (month 30 or May 2016) and ending in the very first month of the 2013 bear run (month 1 or December 2013). The red line represents the ARDL Bull Monthly models predicted value of Bitcoin's low price, and the blue line represents the actual value of Bitcoin's low price. Based on the graph, it is evident that the ARDL Bull Monthly model performed impressively well during the 2013 bear run. This is due to its predicted values proximity to the actual value throughout the 2013 bear run as evident by the red and blue lines as well as its low MAPE of 4.5% which is significantly below the 10% benchmark set by Lewis (1982) in the Journal of Forecasting.

Figure X: ARDL Bull Monthly 2016 Bull Run Out of Sample Back Test Results



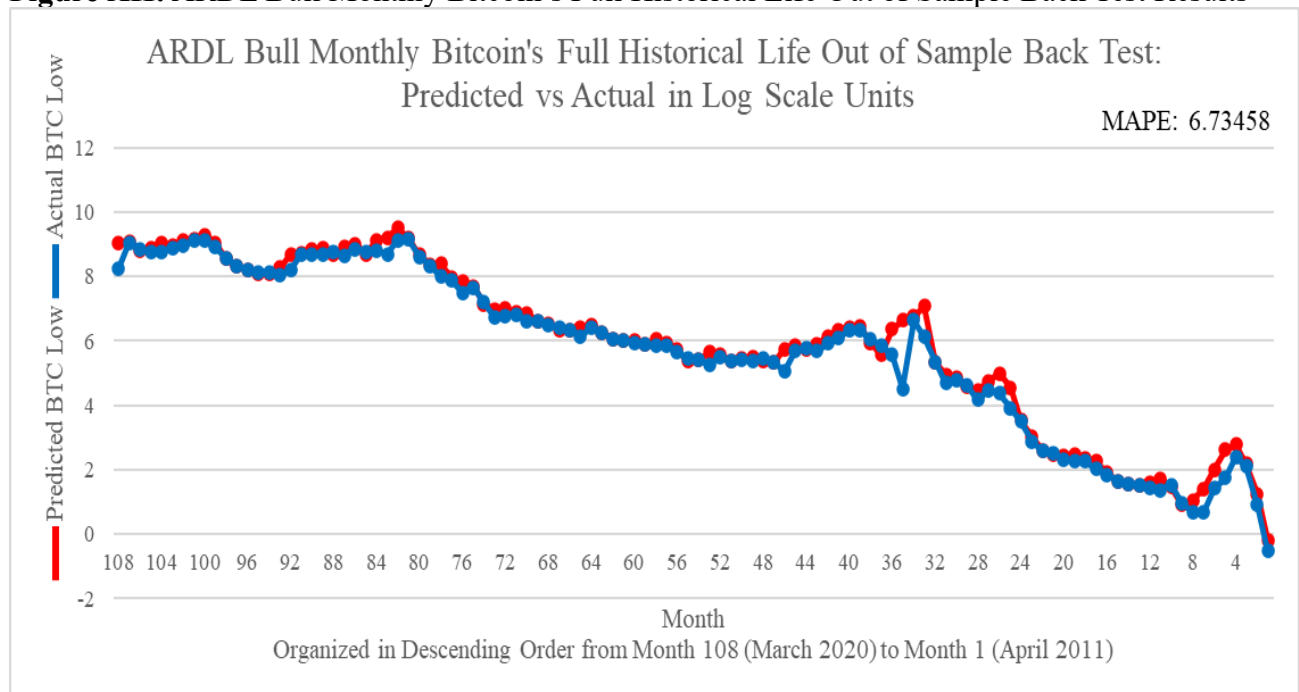
³¹ Figure X provides a visual graphical representation of the 2016 bull run out of sample back test MAPE results of the ARDL Bull Monthly model for predicting Bitcoin's low price. The graph moves from left to right in descending months starting with the very last month of the 2016 bull run (month 20 or December 2017) and ending in the very first month of the 2016 bull run (month 1 or May 2016). The red line represents the ARDL Bull Monthly models predicted value of Bitcoin's low price, and the blue line represents the actual value of Bitcoin's low price. Based on the graph, it is evident that the ARDL Bull Monthly model performed very well during the 2016 bull run. This is due to its predicted values proximity to the actual value throughout the 2016 bull run as evident by the red and blue lines as well as its low MAPE of 1.6% which is significantly below the 10% benchmark set by Lewis (1982) in the Journal of Forecasting.

Figure XI: ARDL Bull Monthly 2017 Bear Run Out of Sample Back Test Results



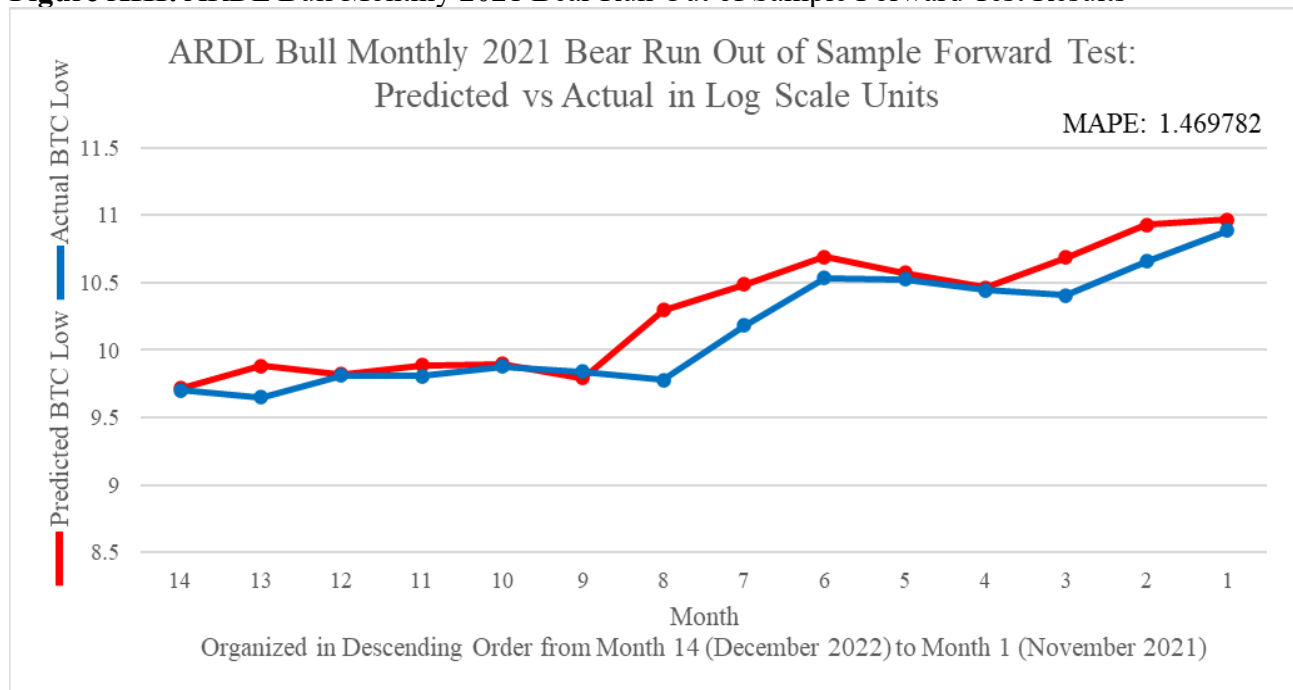
³² Figure XI provides a visual graphical representation of the 2017 bear run out of sample back test MAPE results of the ARDL Bull Monthly model for predicting Bitcoin's low price. The graph moves from left to right in descending months starting with the very last month of the 2017 bear run (month 28 or March 2020) and ending in the very first month of the 2017 bear run (month 1 or December 2017). The red line represents the ARDL Bull Monthly models predicted value of Bitcoin's low price, and the blue line represents the actual value of Bitcoin's low price. Based on the graph, it is evident that the ARDL Bull Monthly model performed very well during the 2017 bear run. This is due to its predicted values proximity to the actual value throughout the 2017 bear run as evident by the red and blue lines as well as its low MAPE of 2.0% which is significantly below the 10% benchmark set by Lewis (1982) in the Journal of Forecasting.

Figure XII: ARDL Bull Monthly Bitcoin's Full Historical Life Out of Sample Back Test Results



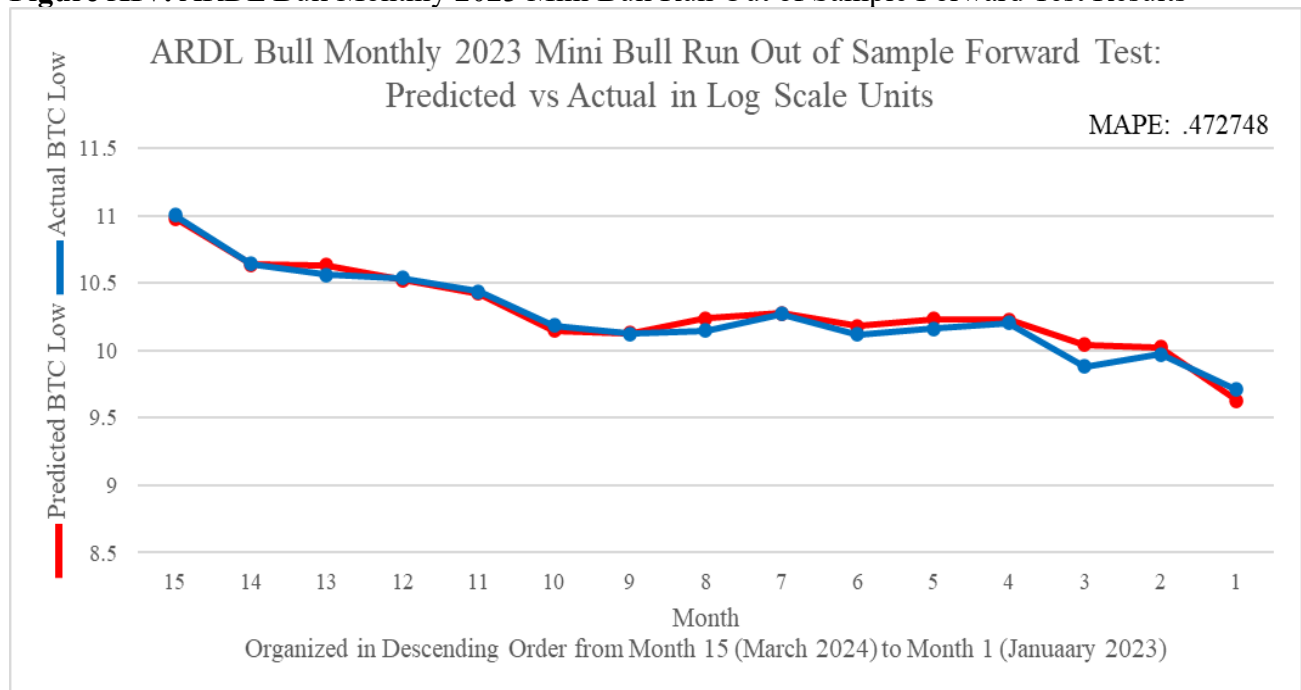
³³ Figure XII provides a visual graphical representation of Bitcoin's full historical life out of sample back test MAPE results of the ARDL Bull Monthly model for predicting Bitcoin's low price. The graph moves from left to right in descending months starting with the last month of Bitcoin's full historical life (month 108 or March 2020) and ending in the first month of Bitcoin's full historical life (month 1 or April 2011). The red line represents the ARDL Bull Monthly models predicted value of Bitcoin's low price, and the blue line represents the actual value of Bitcoin's low price. Based on the graph, it is evident that the ARDL Bull Monthly model performed very well during Bitcoin's full historical life. This is due to its predicted values proximity to the actual value throughout Bitcoin's full historical life as evident by the red and blue lines as well as its low MAPE of 6.7% which is significantly below the 10% benchmark set by Lewis (1982) in the Journal of Forecasting.

Figure XIII: ARDL Bull Monthly 2021 Bear Run Out of Sample Forward Test Results



³⁴ Figure XIII provides a visual graphical representation of the 2021 bear run out of sample forward test MAPE results of the ARDL Bull Monthly model for predicting Bitcoin's low price. The graph moves from left to right in descending months starting with the very last month of the 2021 bear run (month 14 or December 2022) and ending in the very first month of the 2021 bear run (month 1 or November 2021). The red line represents the ARDL Bull Monthly models predicted value of Bitcoin's low price, and the blue line represents the actual value of Bitcoin's low price. Based on the graph, it is evident that the ARDL Bull Monthly model performed very well during the 2021 bear run. This is due to its predicted values proximity to the actual value throughout the 2021 bear run as evident by the red and blue lines as well as its low MAPE of 1.4% which is significantly below the 10% benchmark set by Lewis (1982) in the Journal of Forecasting.

Figure XIV: ARDL Bull Monthly 2023 Mini Bull Run Out of Sample Forward Test Results



³⁵ Figure XIV provides a visual graphical representation of the 2023 mini bull run out of sample forward test MAPE results of the ARDL Bull Monthly model for predicting Bitcoin's low price. The graph moves from left to right in descending months starting with the very last month of the 2023 mini bull run (month 15 or March 2024) and ending in the very first month of the 2023 mini bull run (month 1 or January 2023). The red line represents the ARDL Bull Monthly models predicted value of Bitcoin's low price, and the blue line represents the actual value of Bitcoin's low price. Based on the graph, it is evident that the ARDL Bull Monthly model performed very well during the 2023 mini bull run. This is due to its predicted values proximity to the actual value throughout the 2023 mini bull run as evident by the red and blue lines as well as its low MAPE of 0.47% which is significantly below the 10% benchmark set by Lewis (1982) in the Journal of Forecasting.